

Rank Determination for Time Series Tensor Factor Model Using Correlation Thresholding

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Abstract

We introduce the total mode- k correlation for time series from a tensor factor model, and propose a tuning-parameter free procedure for determining the rank of the core tensor of the common component. The use of correlation information is inspired by Fan et al. (2020), where they use the eigenvalues in the correlation matrix of the data for determining the number of factors in a factor model with independent observations. We consider tensor time series data with serially and cross-sectionally correlated factors and idiosyncratic noise. Two asymptotic regimes are considered for each order of the tensor. The first one is when the order- k dimension, d_k , is growing slower than the sample size T , so that $d_k/T \rightarrow 0$ as $T \rightarrow \infty$. The second one is when $d_k/(Td_{-k}) \rightarrow 0$, where $d_{-k} = \prod_{j \neq k} d_j$ is the product of all dimensions except the k -th. The second regime entails d_k growing faster than T . Theoretical properties of our estimators by total mode- k correlation thresholding (abbreviated as TCorTh) are presented. We compare TCorTh empirically to the eigenvalue-ratio (ER) and information criterion (IC) procedures proposed by Han et al. (2020), where the two procedures are designed for the idiosyncratic noise being serially uncorrelated. A set of financial data is analysed as well.

Key words and phrases. Multilinear rank; total mode- k covariance matrix; mode- k unfolding; k -mode product; eigenvalues thresholding.

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1 Introduction

With the advance of computer power over the past decade, the analysis of large time series has been made ever easier, allowing for more computationally demanding algorithms to be implemented. There is usually a large number of time series in a set of tensor time series data, making it very large dimensional relative to the sample size. This is especially true for finance or economic data for example, when the sample size can be limited by the amount of data available, or we restrict the sample size due to the concern of nonstationarity over a long horizon. To tackle the problem of large dimensionality and small sample size, a tensor factor model serves as an important dimension reduction tool to make empirical analysis possible again and add structures and interpretations to the tensor data at hand. Similar to its factor model counterpart for a panel of time series, we need to find the number of “factors” that is driving the dynamics of all the time series in the tensor data.

A general decomposition of an order- K tensor time series data $\mathcal{X}_t \in \mathbb{R}^{d_1 \times \dots \times d_K}$ is given by

$$\mathcal{X}_t = \mu + \mathcal{C}_t + \mathcal{E}_t, \quad t = 1, \dots, T, \quad (1.1)$$

where $\mu \in \mathbb{R}^{d_1 \times \dots \times d_K}$ is the mean of $\mathcal{X}_t$, $\mathcal{C}_t$ is called the common component and $\mathcal{E}_t$ the idiosyncratic error. The common component $\mathcal{C}_t$ is usually driven by just a small number of “factors”. The exact decomposition of $\mathcal{C}_t$ will be given in model (3.4) in Section 3. When $K = 1$, i.e., $\mathcal{X}_t$ is a vector of observed time series, Bai and Ng (2002) proposed two information criteria for consistently finding the number of factors r . Ahn and Horenstein (2013) find the number of factors r by maximizing the ratio of consecutive eigenvalues of the sample covariance matrix. Both papers exploit the fact that the first r eigenvalues of the sample covariance matrix diverge to infinity, whereas the others remain bounded. Both of them considered an approximate factor model (see Chamberlain and Rothschild (1983) for more discussions on approximate factor models), where the elements in $\mathcal{E}_t$ can be both serially and contemporaneously correlated to a certain extent.

Still with $K = 1$, Lam and Yao (2012) proposed an eigenvalues ratio based estimator for finding the number of factors on a statistical factor model using sample autocovariance matrices, where $\mathcal{E}_t$ is assumed to be serially uncorrelated. Such an assumption entails that all serial information for $\mathcal{X}_t$ is contained in the factors in $\mathcal{C}_t$, which facilitates a simple eigenanalysis for the estimation of the factor model.

For $K > 1$, there are two main decompositions of $\mathcal{C}_t$ for a tensor factor model, namely the CP decomposition and the Tucker decomposition. The latter is model (3.4) in Section 3, while the former will not be discussed in this paper. See for example Wang et al. (2019) for interpretation of factor loading matrices

for a tensor factor model with $K = 2$ (matrix-valued time series), and also the real data applications in Chen et al. (2021) for some interpretations of factor loading matrices with $K > 2$, all under the Tucker decomposition of the common component. Information criteria (IC) and eigenvalues ratio (ER) based methods are extended to a general K in Han et al. (2020) for a statistical tensor factor model under the Tucker decomposition, with all the elements in $\mathcal{E}_t$ being serially uncorrelated. While their methods are also proved to be consistent for the number of factors, penalty functions are needed to be specified for both IC and ER methods, and potential tuning parameters are needed for fine tuning of performance. With more complex structures in a tensor factor model compared to vector time series factor models, tuning parameters can be computationally expensive to optimize the performance of the estimators. Their TOPUP procedure requiring the processing of a potentially huge five-dimensional array (with two dimensions being the product of the original tensor data dimensions) also does not align well with the analysis of high dimensional tensor data with $K \geq 4$. Even if $K \leq 3$, TOPUP is in general quite slow relative to their TIPUP counterpart (see Section 6 for our simulation results). At the same time, it may not be ideal to assume all elements of $\mathcal{E}_t$ to be white noise, i.e., without any serial correlations. Allowing serial correlations in the noise can be especially important when the data measures economic and financial activities, where idiosyncratic errors can exhibit weak serial dependence due to unique company/regional effects for instance.

There are few other attempts to estimate the number of factors in a tensor factor model under Tucker decomposition of the common component with noise. Under this setup, Yokota et al. (2017) carefully writes the tensor data as a matrix first, and use the eigenvalues of the resulting sample covariance matrix to construct a BIC-like criterion for minimization. The noises have to be i.i.d. Gaussian however, greatly reducing the usability of the method. Han et al. (2018) proposed using mixed and L_1 norms to promote sparsity in the estimation of the core tensor, which is over-estimated in the sense that initially (large) prefixed ranks are used in the estimation problem. Then they proposed a criterion for minimization involving the singular values of the unfolded sparsely estimated core tensor (tensor unfolding will be briefly introduced in Section 2). This is a two-step procedure to estimate the ranks, with prefixed inputs needed. Large prefixed ranks means the procedure suffers in computational complexity, while small prefixed ranks may result in underestimation of the ranks themselves. Also, consistency results are not given. Other attempts on rank estimation of a tensor factor model are more commonly done on the tensor rank itself, which is the rank of a CP decomposition. For instance, Shi et al. (2017) proposed a penalization method in estimating the CP rank of a given tensor data in the absence of noise.

In this paper, we propose a tuning parameter-free and one-step procedure for estimating the ranks of the core tensor under the Tucker decomposition of $\mathcal{C}_t$ in model (1.1). Contemporaneous and serial

correlations are allowed in $\mathcal{E}_t$, which is particularly important for economic and financial applications. Our method involves K eigenanalyses of some correlation matrices for finding the K core ranks of an order- K tensor. This is inspired by the tuning parameter-free method in finding the number of factors in a factor model in Fan et al. (2020), where the eigenvalues of the correlation matrix of the data are compared to a known threshold. Since we are thresholding the eigenvalues of the newly defined total model- k correlation matrix, our method is abbreviated as TCorTh. We extend their method to our tensor model setting in (1.1), with time series variables allowed. We also discuss two asymptotic regimes, with almost sure convergence results given over them in two theorems in Section 4. Rates of convergence are spelt out as well.

The rest of the paper is organized as follows. Section 2 introduces all notations and basic manipulations of tensors needed for our presentation, including tensor unfolding and the k -mode product. Section 3 describes the tensor factor model and the assumptions needed for the tensor data. Our rank estimation procedure is then introduced. Section 4 presents the associated theoretical results for our rank estimators TCorTh. Section 6 contains extensive simulation studies comparing TCorTh to the IC and ER-based estimators proposed in Han et al. (2020). A set of financial data is also analyzed using TCorTh and compared with the iTOPUP and iTIPUP used in Han et al. (2020). All proofs of our results are in section 8.

2 Notations and Basic Tensor Manipulations

We briefly introduce the notations and review on tensor manipulations in this section just enough for the presentation of this paper. For more information, please refer to Kolda and Bader (2009).

Let $\mathcal{X} \in \mathbb{R}^{d_1 \times \cdots \times d_K}$ be an order- K tensor. Here K represents the number of dimensions in $\mathcal{X}_t$, also called the number of *modes*. For instance, a vector time series has $K = 1$ while a matrix time series has $K = 2$. If we write $\mathcal{X} = (x_{i_1 \cdots i_K})$, then we define a *mode- k fibre* of $\mathcal{X}$ to be a column vector (of length d_k)

$$(x_{i_1 \cdots i_{k-1}, j, i_{k+1} \cdots i_K})_{j \in [d_k]}, \quad i_\ell \in [d_\ell] \text{ with } \ell \in [K],$$

where we define, for any positive integer n , $[n] := \{1, \dots, n\}$. Hence there are in total $d_{-k} := \prod_{\ell=1, \ell \neq k}^K d_\ell$ number of mode- k fibres for the tensor $\mathcal{X}$. The *mode- k unfolding matrix* $\mathbf{X}_{(k)} \in \mathbb{R}^{d_k \times d_{-k}}$ is then defined to be the matrix containing (in order) all the mode- k fibres of $\mathcal{X}$.

If there is a matrix $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{I_k \times r_k}$, $k \in [K]$, and $\mathcal{F} = (f_{i_1 \cdots i_K}) \in \mathbb{R}^{r_1 \times \cdots \times r_K}$ is an order- K tensor,

then the k -mode product of $\mathcal{F}$ and $\mathbf{A}$, denoted by $\mathcal{F} \times_k \mathbf{A} \in \mathbb{R}^{r_1 \times \cdots \times r_{k-1} \times I_k \times r_{k+1} \times \cdots \times r_K}$, is defined as

$$(\mathcal{F} \times_k \mathbf{A})_{i_1 \cdots i_{k-1}, j, i_{k+1} \cdots i_K} := \sum_{i_k=1}^{r_k} f_{i_1 \cdots i_k \cdots i_K} a_{ji_k}.$$

As an example, consider $K = 2$, so that $\mathcal{F}$ is a matrix. Let $\mathbf{A}_1 \in \mathbb{R}^{I_1 \times r_1}$ and $\mathbf{A}_2 \in \mathbb{R}^{I_2 \times r_2}$. Then the mode-1 fibres of $\mathcal{F}$ are in fact the columns of $\mathcal{F}$, while the mode-2 fibres of $\mathcal{F}$ are the rows of $\mathcal{F}$ (made column vectors). We also have

$$\mathcal{F} \times_1 \mathbf{A}_1 = \mathbf{A}_1 \mathcal{F}, \quad \mathcal{F} \times_2 \mathbf{A}_2 = \mathcal{F} \mathbf{A}_2^\top = (\mathbf{A}_2 \mathcal{F}^\top)^\top.$$

Note that the two unfolded matrices (mode-1 and mode-2 respectively) are $\mathbf{F}_{(1)} = \mathcal{F}$ and $\mathbf{F}_{(2)} = \mathcal{F}^\top$, hence the above also means that

$$\mathcal{F} \times_1 \mathbf{A}_1 = \mathbf{A}_1 \mathbf{F}_{(1)}, \quad \mathcal{F} \times_2 \mathbf{A}_2 = (\mathbf{A}_2 \mathbf{F}_{(2)})^\top.$$

In general, to calculate $\mathcal{F} \times_k \mathbf{A}$, we find the mode- k unfolding matrix $\mathbf{F}_{(k)}$ first and then calculate $\mathbf{A} \mathbf{F}_{(k)}$, which contains all mode- k fibres of $\mathcal{F}$ being pre-multiplied by $\mathbf{A}$. Then we put the columns in $\mathbf{A} \mathbf{F}_{(k)}$ back into the original shape of the tensor $\mathcal{F}$. Hence for $K = 2$, to put the columns in $\mathbf{A} \mathbf{F}_{(2)}$ back into the original orientation of the tensor (rows), we take transpose of it, so that $\mathcal{F} \times_2 \mathbf{A} = (\mathbf{A} \mathbf{F}_{(2)})^\top$. The order of distinct mode products does not matter, in the sense that for $i \neq j$,

$$\mathcal{F} \times_i \mathbf{A}_i \times_j \mathbf{A}_j = \mathcal{F} \times_j \mathbf{A}_j \times_i \mathbf{A}_i.$$

Finally, if $\mathcal{C} = \mathcal{F} \times_1 \mathbf{A}_1 \times_2 \cdots \times_K \mathbf{A}_K$, then we have the formula

$$\mathbf{C}_{(k)} = \mathbf{A}_k \mathbf{F}_{(k)} \mathbf{A}_{-k}^\top, \tag{2.2}$$

where $\otimes$ is the Kronecker product, $\mathbf{A}_{-k} := \mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_{k+1} \otimes \mathbf{A}_{k-1} \otimes \cdots \otimes \mathbf{A}_1$, and $\mathbf{C}_{(k)}$ is the mode- k unfolding of $\mathcal{C}$. Hence, denote by $\text{vec}(\mathcal{X})$ the vectorization of $\mathcal{X}$, which is to stack all mode-1 fibres of $\mathcal{X}$ into a column vector, we have

$$\text{vec}(\mathbf{C}_{(k)}) = (\mathbf{A}_{-k} \otimes \mathbf{A}_k) \text{vec}(\mathbf{F}_{(k)}).$$

In particular,

$$\text{vec}(\mathcal{C}) = \text{vec}(\mathbf{C}_{(1)}) = (\mathbf{A}_{-1} \otimes \mathbf{A}_1) \text{vec}(\mathbf{F}_{(1)}) = (\mathbf{A}_K \otimes \cdots \otimes \mathbf{A}_1) \text{vec}(\mathcal{F}). \tag{2.3}$$

3 Tensor Factor Model and Rank Estimators

Consider an observed order- K tensor time series $\mathcal{X}_t \in \mathbb{R}^{d_1 \times \dots \times d_K}$, $t \in [T]$. We consider a decomposition in (1.1), with the Tucker decomposition of the common component $\mathcal{C}_t$ as follows:

$$\mathcal{X}_t = \mu + \mathcal{C}_t + \mathcal{E}_t = \mu + \mathcal{F}_t \times_1 \mathbf{A}_1 \times_2 \dots \times_K \mathbf{A}_K + \mathcal{E}_t, \quad (3.4)$$

where $\mathcal{F}_t \in \mathbb{R}^{r_1 \times \dots \times r_K}$ is called a core tensor, and $\mathbf{A}_k \in \mathbb{R}^{d_k \times r_k}$, $k \in [K]$, is called a mode- k factor loading matrix. Decomposition of tensor data like (3.4) is also considered in Yokota et al. (2017) and De Lathauwer et al. (2000) in a non-time series context, for instance. The Tucker decomposition of the common component $\mathcal{C}_t \in \mathbb{R}^{d_1 \times \dots \times d_K}$ into $\mathcal{F}_t \times_1 \mathbf{A}_1 \times_2 \dots \times_K \mathbf{A}_K$ achieves vast dimension reductions. Instead of the need to model $d := \prod_{\ell=1}^K d_\ell$ time series, where d can be extremely large, we only need to model $r := \prod_{\ell=1}^K r_\ell$ time series, which is typically much smaller than d since each r_k is assumed to be much smaller than d_k for each k . The dynamics of the time series in $\mathcal{X}_t$ are then driven by the small number of time series in $\mathcal{F}_t$, called the *factors*, and hence co-movements of time series in $\mathcal{X}_t$ are possible. The matrices $\mathbf{A}_1, \dots, \mathbf{A}_K$ are governing how each factor affects each time series in $\mathcal{X}_t$, and are called the *factor loading matrices*.

Using the vectorization formula in (2.3), we can easily see that model (3.4) is equivalent to

$$\text{vec}(\mathcal{X}_t) = \text{vec}(\mu) + (\mathbf{A}_K \otimes \dots \otimes \mathbf{A}_1) \text{vec}(\mathcal{C}_t) + \text{vec}(\mathcal{E}_t), \quad t \in [T],$$

which is a factor model for the vector time series $\text{vec}(\mathcal{X}_t)$ with factors $\text{vec}(\mathcal{C}_t)$ and factor loading matrix $\mathbf{A}_K \otimes \dots \otimes \mathbf{A}_1$. If we do not treat $\text{vec}(\mathcal{X}_t)$ as from a tensor, then the factor loading matrix from a general factor model for $\text{vec}(\mathcal{X}_t)$ will have $dr = \prod_{\ell=1}^K d_\ell r_\ell$ unknown entries to be estimated. Whereas for the model above, the number of unknown parameters from $\mathbf{A}_1$ to $\mathbf{A}_K$ is just $\sum_{\ell=1}^K d_\ell r_\ell$, a huge dimension reduction compared to the general factor model for $\text{vec}(\mathcal{X}_t)$.

3.1 Main assumptions

We present the main assumptions for the tensor factor model in this section. Using mode- k unfolding of $\mathcal{X}_t$, we write model (3.4) as

$$\begin{aligned} \text{mat}_k(\mathcal{X}_t) &= \text{mat}_k(\mu) + \text{mat}_k(\mathcal{C}_t) + \text{mat}_k(\mathcal{E}_t) \\ &= \text{mat}_k(\mu) + \mathbf{A}_k \text{mat}_k(\mathcal{F}_t) \mathbf{A}_{-k}^\top + \text{mat}_k(\mathcal{E}_t), \quad t \in [T], \quad k \in [K], \end{aligned} \quad (3.5)$$

where $\text{mat}_k(\mathcal{X})$ extracts the mode- k unfolding matrix of the tensor $\mathcal{X}$, and $\mathbf{A}_{-k}$ is as defined in (2.2).

3.1.1 Assumptions on the errors

We present assumptions (E1) - (E2) below with explanations.

(E1) (Decomposition of error) *Assume that*

$$\begin{aligned} \text{mat}_k(\mathcal{E}_t) &= (\boldsymbol{\xi}_{t,1}^{(k)}, \dots, \boldsymbol{\xi}_{t,d_k}^{(k)}), \quad \text{where} \\ \boldsymbol{\xi}_{t,\ell}^{(k)} &:= \boldsymbol{\Psi}_\ell^{(k)} \mathbf{e}_t^{(k)} + (\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)})^{1/2} \boldsymbol{\epsilon}_{t,\ell}^{(k)}, \end{aligned} \quad (3.6)$$

with $E(\mathbf{e}_t^{(k)}) = \mathbf{0}$, $E(\boldsymbol{\xi}_{t,\ell}^{(k)}) = \mathbf{0}$, $\mathbf{e}_t^{(k)} \in \mathbb{R}^{r_e}$ independent of $\boldsymbol{\epsilon}_{s,\ell}^{(k)}$, $\boldsymbol{\epsilon}_{t,\ell}^{(k)}$ independent of $\boldsymbol{\epsilon}_{t,m}^{(k)}$ for $\ell \neq m$, $\text{var}(\mathbf{e}_t^{(k)}) = \mathbf{I}_{r_e}$ and $\text{var}(\boldsymbol{\epsilon}_{t,\ell}^{(k)}) = \mathbf{I}_{d_k}$ for each $s, t \in [T]$, $\ell, m \in [d_k]$, $k \in [K]$. Also, each $\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)}$ has non-vanishing diagonals with $\text{tr}(\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)}) = O(d_k)$, where $\text{tr}(\cdot)$ is the trace of a square matrix.

Hence with (E1), each mode- k fibre of the error tensor $\mathcal{E}_t$ is a sum of two independent parts. The first part is $\boldsymbol{\Psi}_\ell^{(k)} \mathbf{e}_t^{(k)}$, which is similar to a common component in a factor model because $\mathbf{e}_t^{(k)}$ is independent of the index ℓ , so that it appears in every mode- k fibre at time t . However, we assume $\boldsymbol{\Psi}_\ell^{(k)}$ is sparse or approximately sparse, meaning that this common component is too weak to be detected. See more detailed assumptions in Section 4. We allow the existence of this part in our model because this promotes (weak) cross-correlations among the fibres: For $\ell, m \in [d_k]$,

$$\text{cov}(\boldsymbol{\xi}_{t,\ell}^{(k)}, \boldsymbol{\xi}_{t,m}^{(k)}) = \boldsymbol{\Psi}_\ell^{(k)} \boldsymbol{\Psi}_m^{(k)\top}.$$

For instance, if $\mathcal{X}_t \in \mathbb{R}^{10 \times 10}$ is a matrix-valued time series ($K = 2$) of 100 asset returns, with rows denoting 10 different size levels and columns denoting 10 different book-to-market levels, then the error tensor series $\mathcal{E}_t$ can exhibit weak cross-correlations, especially if the assets are all within the same market and some are closely related (two assets with the same parent company, for instance). The inclusion of the term $\boldsymbol{\Psi}_\ell^{(k)} \mathbf{e}_t^{(k)}$ for each mode- k fibre facilitates such weak cross-correlations. The last part $\text{tr}(\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)}) = O(d_k)$ ensures that the error variances are all constants, or only finitely many of them are “large” with magnitude $O(d_k)$.

(E2) (Time series) *The elements in $\mathbf{e}_t^{(k)} = (e_{t,j}^{(k)})$ and $\boldsymbol{\epsilon}_{t,\ell}^{(k)} = (\epsilon_{t,\ell,j}^{(k)})$ are following weakly stationary general linear processes, such that with $\ell \in [d_k]$, $t \in [T]$ and $k \in [K]$,*

$$\begin{aligned} e_{t,j}^{(k)} &= \sum_{q \geq 0} a_{e,q} z_{e,t-q,j}^{(k)}, \quad j \in [r_e], \\ \epsilon_{t,\ell,j}^{(k)} &= \sum_{q \geq 0} a_{\epsilon,q} z_{\epsilon,t-q,\ell,j}^{(k)}, \quad j \in [d_k], \end{aligned} \quad (3.7)$$

where the coefficients $a_{e,q}$ and $a_{\epsilon,q}$ are such that $\sum_{q \geq 0} a_{e,q}^2 = \sum_{q \geq 0} a_{\epsilon,q}^2 = 1$. For each $k \in [K]$, the series of random variables $\{z_{e,t,j}^{(k)}\}$ and $\{z_{\epsilon,t,\ell,j}^{(k)}\}$ are independent of each other, with i.i.d. elements having mean 0 and variance 1.

The rate at which the $a_{e,j}$'s and $a_{\epsilon,j}$'s are going to 0, and the moment assumptions for the $z_{e,t,j}^{(k)}$'s and $z_{\epsilon,t,\ell,j}^{(k)}$'s are presented in details in Section 4. With this assumption, the error variables are serially correlated in general. Together with (E1), (weak) serial and cross-sectional dependence within and among fibres are allowed for the errors.

3.1.2 Assumptions on the factors

Similar to (E2), the factors in $\mathcal{F}_t$ are assumed to follow general linear processes.

(F1) Let $\mathbf{f}_{t,\ell}^{(k)} = (f_{t,\ell,j}^{(k)})$ be the ℓ -th column vector in $\text{mat}_k(\mathcal{F}_t)$, $\ell \in [r_{-k}]$, where $r_{-k} := \prod_{\ell \neq k} r_\ell$. We assume that $\text{var}(\mathbf{f}_{t,\ell}^{(k)}) = \mathbf{I}_{r_k}$ (the identity matrix with size r_k), and $\text{cov}(\mathbf{f}_{t,\ell_1}^{(k)}, \mathbf{f}_{t,\ell_2}^{(k)}) = \mathbf{0}$ for $\ell_1 \neq \ell_2$.

Then we can write

$$f_{t,\ell,j}^{(k)} = \sum_{q \geq 0} a_{f,q} z_{f,t-q,\ell,j}^{(k)}, \quad j \in [r_k], \quad (3.8)$$

where we have $\sum_{q \geq 0} a_{f,q}^2 = 1$. For each $k \in [K]$, the series of random variables $\{z_{f,t,\ell,j}^{(k)}\}$ has i.i.d. elements having zero mean and variance 1.

Note the series of coefficients $\{a_{e,q}\}$, $\{a_{\epsilon,q}\}$ and $\{a_{f,q}\}$ are not necessarily equal.

3.1.3 Assumptions on the model parameters

We present the assumptions needed for the factor loading matrices $\mathbf{A}_k$, $k \in [K]$, and other model parameters. Part of the assumptions are in fact parallel to (12) in Fan et al. (2020), and can be considered an extension to theirs from factor model to tensor factor model.

(M1) Define the total mode- k covariance matrix for $\mathcal{X}_t$ to be

$$\Sigma^{(k)} := E(\text{mat}_k(\mathcal{X}_t - E(\mathcal{X}_t)) \text{mat}_k(\mathcal{X}_t - E(\mathcal{X}_t))^T). \quad (3.9)$$

Then we assume that, for $k \in [K]$, $\mathbf{A}_k$ is of full rank. And there exist constants $c, C > 0$ and $\delta_1 > \delta_2$

with $r_k = o(d_k^{\delta_1 - \delta_2})$, $r = o(T)$ and $r_e = o(d_k)$, such that as $d_k \rightarrow \infty$,

$$\begin{aligned} \|\mathbf{A}_{-k}\|_F^2 \cdot \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|^2 &\geq C d_k^{\delta_1}, \\ c d_k^{\delta_2} &\leq \|\mathbf{A}_k^T \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\| \cdot \|\{\mathbf{A}_k^T \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\}^{-1}\| \leq C d_k^{\delta_2}, \\ \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} (\boldsymbol{\Sigma}_\epsilon^{(k)} + \boldsymbol{\Psi}^{(k)} \boldsymbol{\Psi}^{(k)T})\| &\leq 1, \quad \sum_{j=1}^{d_k} (\|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \boldsymbol{\Sigma}_{\epsilon,j}^{(k)}\| + \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \boldsymbol{\Psi}_j^{(k)} \boldsymbol{\Psi}_j^{(k)T}\|) < \infty, \end{aligned} \quad (3.10)$$

where $\boldsymbol{\Sigma}_\epsilon^{(k)} := \sum_{\ell=1}^{d-k} \boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)}$, $\boldsymbol{\Psi}^{(k)} := (\boldsymbol{\Psi}_1^{(k)}, \dots, \boldsymbol{\Psi}_{d-k}^{(k)})$, and $\|\cdot\|_F$ denotes the Frobenius norm.

From model (3.5) and Assumption (E1) and (F1), we can easily show that (proof omitted)

$$\boldsymbol{\Sigma}^{(k)} = \|\mathbf{A}_{-k}\|_F^2 \mathbf{A}_k \mathbf{A}_k^T + \boldsymbol{\Sigma}_\epsilon^{(k)} + \boldsymbol{\Psi}^{(k)} \boldsymbol{\Psi}^{(k)T}. \quad (3.11)$$

Consider, WLOG, $\boldsymbol{\Sigma}_{\epsilon,\ell}^{(k)} = \mathbf{I}_{d_k}$, so that $\boldsymbol{\Sigma}_\epsilon^{(k)} = d_{-k} \mathbf{I}_{d_k}$. For ease of presentation, suppose the weak common factors in (3.6) is zero, so that $\boldsymbol{\Psi}^{(k)} = \mathbf{0}$. Then we can see immediately that the third condition in (3.10) is automatically satisfied when each $\boldsymbol{\Sigma}_{\epsilon,j}^{(k)}$ is well-conditioned. In general, as long as $\boldsymbol{\Sigma}_\epsilon^{(k)}$ and $\boldsymbol{\Psi}^{(k)}$ are sparse enough with the majority of $\boldsymbol{\Sigma}_{\epsilon,j}^{(k)}$'s well-conditioned, the condition will also be satisfied.

Furthermore, consider that each $\mathbf{A}_k \in \mathbb{R}^{d_k \times r_k}$ has $\|\mathbf{A}_k\|_F^2$ with order at least d_k , with notation $\|\mathbf{A}_k\|_F^2 \succeq d_k$. Then $\|\mathbf{A}_{-k}\|_F^2 = \prod_{\ell \neq k} \|\mathbf{A}_\ell\|_F^2 \succeq d_{-k}$. The first condition in (3.10) is then satisfied with $\delta_1 = 1$. And, as long as the columns in $\mathbf{A}_k$ are not asymptotically linearly dependent, so that $d_k^{-1} \mathbf{A}_k^T \mathbf{A}_k$ is a full rank matrix with a non-diverging condition number (or approaches a such a matrix in the case that the elements in $\mathbf{A}_k$ are random), then the second condition in (3.10) is satisfied with $\delta_2 = 0$.

The importance of Assumption (M1) is the following.

Theorem 1. *Let Assumption (E1), (F1) and (M1) hold. Define the total mode- k correlation matrix*

$$\mathbf{R}^{(k)} := \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \boldsymbol{\Sigma}^{(k)} \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2}. \quad (3.12)$$

Then we have $\lambda_j(\mathbf{R}^{(k)}) > 1$ for $j \in [r_k]$, while $\lambda_j(\mathbf{R}^{(k)}) \leq 1$ for $j = r_k + 1, \dots, d_k$, where $\lambda_j(\cdot)$ denotes the j -th largest eigenvalue.

The proof is relegated to Section 8. This theorem is in parallel to Theorem 1 of Fan et al. (2020), and paves the ways to using the eigenvalues of the estimated total mode- k correlation matrix for estimating r_k for each $k \in [K]$, since Theorem 1 implies that

$$r_k = \max\{j : \lambda_j(\mathbf{R}^{(k)}) > 1, j \in [d_k]\}.$$

3.2 Rank estimators

With the results in Theorem 1, we propose to estimate r_k for each $k \in [K]$ by

$$\hat{r}_k := \max\{j : \lambda_j(\hat{\mathbf{R}}^{(k)}) > 1, j \in [d_k]\}, \quad (3.13)$$

where

$$\begin{aligned} \hat{\mathbf{R}}^{(k)} &:= \text{diag}(\hat{\Sigma}^{(k)})^{-1/2} \hat{\Sigma}^{(k)} \text{diag}(\Sigma^{(k)})^{-1/2}, \quad \text{with} \\ \hat{\Sigma}^{(k)} &:= T^{-1} \sum_{t=1}^T \text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}}) \text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}})^\top, \quad \bar{\mathcal{X}} := T^{-1} \sum_{t=1}^T \mathcal{X}_t. \end{aligned} \quad (3.14)$$

Fan et al. (2020) considers the estimator of the number of factors r for a factor model:

$$\hat{r} = \max\{j : \lambda_j(\hat{\mathbf{R}}) > 1 + \sqrt{p/(n-1)}, j \in [p]\},$$

where $\hat{\mathbf{R}}$ is the correlation matrix estimator from data, r is the number of factors, p is the dimension of the observations and n is the sample size. The threshold $1 + \sqrt{p/(n-1)}$ is due to the consideration that p can be growing with n together in the sense that $p/n \rightarrow c > 0$ using random matrix theory.

In this paper, while we consider the case $d_k/T \rightarrow 0$, we also consider $d_k/(Td_k) \rightarrow 0$ under a more detailed assumption on $\Sigma_{\epsilon, \ell}^{(k)}$ for each $\ell \in [d_k]$, to be discussed in Section 4. The latter allows d_k to be growing even faster than T , since we are also borrowing information from samples from other fibres (d_k of them in total). Also, this latter case is usually well satisfied in practice, especially for financial and economic applications. For instance, Wang et al. (2019) considers a data set of 10×10 monthly return series over 624 months. Hence $K = 2$, $d_1 = d_2 = 10$, $T = 624$. While $d_k/T = 10/624$ is already quite small, $d_k/(Td_k) = 10/(624 \cdot 10) = 1/624$ is even smaller, and falls quite well within the regime $d_k/(Td_k) \rightarrow 0$ in practice. Chen et al. (2021) considers a set of monthly multi-category import-export network data of 22 countries and 15 product categories over 84 months, resulting in $T = 84$ observed order-3 tensors each with size $22 \times 22 \times 15$. While d_k/T may not be small enough, $d_k/(Td_k) \leq 1/1260$ for each $k = 1, 2, 3$, which is small enough in practice to assume $d_k/(Td_k) \rightarrow 0$.

4 Theoretical Results

We first present an assumption on $\Sigma_{\epsilon, \ell}^{(k)}$ for $\ell \in [d_k]$ and $k \in [K]$ so that the regime $d_k/(Td_k) \rightarrow 0$ can be considered.

(E3) For each $\ell \in [d_{-k}]$ and $k \in [K]$, let

$$(\boldsymbol{\Sigma}_{\epsilon, \ell}^{(k)})^{1/2} = w_\ell^{(k)} \text{diag}(\sigma_1^{(k)}, \dots, \sigma_{d_k}^{(k)}), \quad (4.15)$$

where the $w_\ell^{(k)}$'s and $\sigma_j^{(k)}$'s satisfy $0 < c < w_\ell^{(k)}, \sigma_j^{(k)} < C < \infty$ uniformly as $d_k, d_{-k} \rightarrow \infty$ for some constants $c, C > 0$.

Under this assumption, the relative ratios of variances within fibres $\boldsymbol{\xi}_{t, \ell}^{(k)}$ are constant across all mode- k fibres, except for those with $\boldsymbol{\Psi}_\ell^{(k)} \neq \mathbf{0}$ (see model (3.6)). For instance, with $K = 2$, we have $w_\ell^{(2)} = \sigma_\ell^{(1)}$ for $\ell \in [d_1]$ and $\sigma_\ell^{(2)} = w_\ell^{(1)}$ for $\ell \in [d_2]$.

To present our theoretical results, consider the following more technical assumptions:

(R1) The elements in $\{z_{f, t, \ell, j}^{(k)}\}$ from Assumption (F1), and those in $\{z_{e, t, j}^{(k)}\}$ and $\{z_{\epsilon, t, \ell, j}^{(k)}\}$ from Assumption (E2) are sub-Gaussian, unless we respectively have $r^3 = o(T)$, $r_e^3 = o(T)$ and $d_k^3 = o(T)$ for $k \in [K]$, in which case we only assume the respective elements have uniformly finite fourth order moments. Also, $T^c = O(d_k)$ for each $k \in [K]$ and $K = o(T^c / \log T)$ for some constant $c > 0$.

(R2) For a positive integer m , let $\mathbf{A}_{f, T} \in \mathbb{R}^{(m+1)T \times T}$ be defined as $\mathbf{A}_{f, T} := (\mathbf{a}_{f, 1}, \dots, \mathbf{a}_{f, T})$, where

$$\mathbf{a}_{f, t} := (\mathbf{0}_{t-1}^\top, a_{f, mT}, a_{f, mT-1}, \dots, a_{f, 0}, \mathbf{0}_{T-t}^\top)^\top, \quad t \in [T],$$

with $\mathbf{0}_j$ being a column vector of j zeros and the $a_{f, q}$'s are from Assumption (F1). Define $\mathbf{A}_{e, T}$ and $\mathbf{A}_{\epsilon, T}$ similarly using coefficients from $\{a_{e, q}\}$ and $\{a_{\epsilon, q}\}$ respectively from Assumption (E2). Then we assume that (with $\mathbf{A}$ can be either $\mathbf{A}_{f, T}, \mathbf{A}_{e, T}$ or $\mathbf{A}_{\epsilon, T}$) $\|\mathbf{A}\|$ is uniformly bounded above, and

$$\frac{1}{T} \text{tr}(\mathbf{A}^\top \mathbf{A}) = 1 - o(T^{-3}d^{-2}), \quad \frac{1}{T} \text{tr}(\mathbf{A}^\top \mathbf{A})^2 \rightarrow a, \quad \frac{1}{T^2} \mathbf{1}_T^\top (\mathbf{A}^\top \mathbf{A})^2 \mathbf{1}_T \rightarrow b, \quad \frac{1}{T^{3/2}} \mathbf{1}_T^\top \mathbf{A}^\top \mathbf{A} \mathbf{1}_T \rightarrow c,$$

where $\mathbf{1}_T$ is a column vector of T ones, and the constant a can be a_f, a_e and a_ϵ for $\mathbf{A} = \mathbf{A}_{f, T}, \mathbf{A}_{e, T}$ and $\mathbf{A}_{\epsilon, T}$ respectively. Similarly for constants b and c . Also for a small constant $\delta > 0$,

$$T^{2\delta} d_k^{1+3\delta_2} r_k d_{-k}^{-1}, \quad d_k^{3\delta_2} d_{-k}^{-1} \sum_{j=1}^{d_k} \|\boldsymbol{\Psi}_j^{(k)}\|^2 = o(1).$$

(R2P) Same as (R2), except $T^{2\delta} d_k^{1+3\delta_2} r_k d_{-k}^{-1} = o(1)$ is replaced by $d_k^{1+3\delta_2} r_k / (T d_{-k}) = o(1)$.

(R1)' Same as (R1), except that $d_k^3 = o(T)$ is replaced by $d_k^3 = o(T d_{-k})$ for $k \in [K]$.

(R2)' Same as (R2), except that $T^{-2}\mathbf{1}_T^T \mathbf{A}_{\epsilon,T}^T \mathbf{A}_{\epsilon,T} \mathbf{1}_T = O(T^{-1})$ and

$$d_k^{3\delta_2} \min\{T^{2\delta} d_k r_k d_{-k}^{-1}, (d_k/(T d_{-k}))^{1/2}\}, \quad d_k^{3\delta_2} d_{-k}^{-1} \sum_{j=1}^{d_{-k}} \|\Psi_j^{(k)}\|^2 = o(1).$$

(R2P)' Same as (R2)', except $d_k^{3\delta_2} \min\{T^{2\delta} d_k r_k d_{-k}^{-1}, (d_k/(T d_{-k}))^{1/2}\} = o(1)$ is replaced by

$$d_k^{3\delta_2} \min\{d_k r_k / (T d_{-k}), (d_k/(T d_{-k}))^{1/2}\} = o(1).$$

Assumption (R1) allows us to use random matrix theory (Theorem 2.8 of Wang and Paul (2014) to be precise) to analyze the eigenvalues of a sample covariance matrix, and the sub-Gaussian condition can be relaxed when the latter rate conditions in (R1) are satisfied. (R1)' is to be coupled with (E3) when (4.15) is valid. For Assumption (R2), consider a truncated linear process $\{y_t\}_{t \in [T]}$, and the original process $\{\tilde{y}_t\}_{t \in [T]}$,

$$\tilde{y}_t = \sum_{q \geq 0} a_q z_{t-q}, \quad y_t = \sum_{q=0}^{mT} a_q z_{t-q}, \quad \text{with } \text{var}(\tilde{y}_t) = 1,$$

where $\{z_t\}$ is a sequence of i.i.d. random variables. Construct the matrix $\mathbf{A}$ using $\{a_q\}$ similar to those in Assumption (R2). Then $\mathbf{A}^T \mathbf{A}$ contains the variance of $\{y_t\}$ on the diagonal, and lag- k autocovariance on the k -th off-diagonal. The rates in (R2) (and (R2)', to be coupled with (E3)) are then controlling how fast the a_q 's are going to 0, and how much serial dependence between the y_t 's are allowed. In particular, general linear processes with absolutely summable autocovariance sequence, short range dependent processes like ARMA models, satisfy the assumption. The rate requirements in (R2P) and (R2P)' are weaker than (R2) and (R2)' respectively, and are required for convergence in probability for the eigenvalues of $\hat{\mathbf{R}}^{(k)}$. We are now ready to present the main theorems of the paper.

Theorem 2. *Let Assumption (E1), (E2), (F1), (M1), (R1) and (R2) hold. Suppose there is a small constant $\delta > 0$ such that $a_T(\delta_2), a'_T(\delta_2) = o(1)$ as $T, d_k \rightarrow \infty$, where*

$$a_T(\delta_2) := d_k^{\delta_2} \sqrt{\frac{r}{T}} + \sqrt{\frac{d_k}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_{-k}} \|\Psi_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} + \frac{T^\delta d_k^{1/2}}{d_{-k}^{1/2}} \left(d_k^{\delta_2/2} r_k^{1/2} + \frac{\|\Psi^{(k)}\|_F}{d_{-k}^{1/2}} \right),$$

$$a'_T(\delta_2) := d_k^{\delta_2} \sqrt{\frac{r}{T}} + \sqrt{\frac{d_k}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_{-k}} \|\Psi_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} + \frac{d_k^{1/2}}{T^{1/2} d_{-k}^{1/2}} \left(d_k^{\delta_2/2} r_k^{1/2} + \frac{\|\Psi^{(k)}\|_F}{d_{-k}^{1/2}} \right).$$

Then for $k \in [K]$,

$$\lambda_j(\hat{\mathbf{R}}^{(k)}) = \begin{cases} \lambda_j(\mathbf{R}^{(k)})(1 + O_{a.s.}(a_T(\delta_2))), & j \in [r_k]; \\ \lambda_j(\mathbf{R}^{(k)}) + O_{a.s.}(a_T(0)), & j \in [d_k] \setminus [r_k]. \end{cases}$$

We also have, for each $k \in [K]$, almost surely,

$$\widehat{r}_k = \max\{j : \lambda_j(\widehat{\mathbf{R}}^{(k)}) > 1, j \in [d_k]\} = r_k.$$

If (R2) is replaced by (R2P), then almost sure convergence becomes convergence in probability, and the convergence rates in $\lambda_j(\widehat{\mathbf{R}}^{(k)})$ become $a'_T(\delta_2)$ and $a'_T(0)$ respectively for $j \in [r_k]$ and $j \in [d_k] \setminus [r_k]$. And, $\widehat{r}_k = r_k$ in probability for each $k \in [K]$.

Remark 1. It is clear that $d_k/T = o(1)$ is needed for $a_T(\delta_2), a'_T(\delta_2) = o(1)$.

To understand better the rates above, suppose $K = 3$, $d_1 = d_2 = d_3 = T^{2/3}$ and each r_k is finite. Suppose also each factor loading matrix $\mathbf{A}_k$ has all factors as pervasive factors, i.e., each column in each $\mathbf{A}_k$ has $O(d_k)$ non-zeros (so that $\delta_1 = 1, \delta_2 = 0$ in (3.10)). Let each $\boldsymbol{\Sigma}_{\epsilon, \ell}^{(k)}$, $\ell \in [d_{-k}]$, be sparse with $\|\boldsymbol{\Sigma}_{\epsilon, \ell}^{(k)}\| < C < \infty$ for some constant $C > 0$. Finally, let $r_e = T^{1/2}$, and each $\boldsymbol{\Psi}_\ell^{(k)}$ is zero except a finite number of them with $\|\boldsymbol{\Psi}_\ell^{(k)}\|_F^2$ being finite. Then for each $k \in [K]$,

$$\lambda_j(\widehat{\mathbf{R}}^{(k)}) = \begin{cases} \lambda_j(\mathbf{R}^{(k)})(1 + O_{a.s.}(T^{-1/6})), & j \in [r_k]; \\ \lambda_j(\mathbf{R}^{(k)}) + O_{a.s.}(T^{-1/6}), & j \in [d_k] \setminus [r_k]. \end{cases}$$

Note also that if $K = 2$, i.e., the data is matrix-valued, then d_k/d_{-k} is either d_1/d_2 ($k = 1$) or d_2/d_1 ($k = 2$), and so it cannot be that $d_k/d_{-k} \rightarrow 0$ for both $k = 1, 2$. Hence the term $T^\delta d_k^{\delta_2/2} r_k^{1/2} (d_k/d_{-k})^{1/2}$ cannot be going to 0 for both $k = 1, 2$, and almost sure convergence of $\widehat{r}_k$ for both $k = 1, 2$ is not ensured. Convergence in probability is still possible though.

Theorem 3. *Let Assumption (E1), (E2), (E3), (F1), (M1), (R1)' and (R2)' hold. Suppose there is a small constant $\delta > 0$ such that $b_T(\delta_2), b'_T(\delta_2) = o(1)$ as $T, d_k \rightarrow \infty$, where*

$$\begin{aligned} b_T(\delta_2) &:= d_k^{\delta_2} \sqrt{\frac{r}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_{-k}} \|\boldsymbol{\Psi}_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} \\ &+ d_k^{\delta_2/2} \min \left\{ \frac{T^\delta d_k^{1/2} r_k^{1/2}}{d_{-k}^{1/2}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} + \min \left\{ \frac{T^\delta d_k^{1/2} \|\boldsymbol{\Psi}^{(k)}\|_F}{d_{-k}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\}, \end{aligned}$$

and $b'_T(\delta_2)$ has T^δ replaced by $T^{-1/2}$. Then for $k \in [K]$,

$$\lambda_j(\widehat{\mathbf{R}}^{(k)}) = \begin{cases} \lambda_j(\mathbf{R}^{(k)})(1 + O_{a.s.}(b_T(\delta_2))), & j \in [r_k]; \\ \lambda_j(\mathbf{R}^{(k)}) + O_{a.s.}(b_T(0)), & j \in [d_k] \setminus [r_k]. \end{cases}$$

We also have, for each $k \in [K]$, almost surely,

$$\hat{r}_k = \max\{j : \lambda_j(\hat{\mathbf{R}}^{(k)}) > 1, j \in [d_k]\} = r_k.$$

If $(R2)'$ is replaced by $(R2P)'$, then almost sure convergence becomes convergence in probability, and the convergence rates in $\lambda_j(\hat{\mathbf{R}}^{(k)})$ become $b'_T(\delta_2)$ and $b'_T(0)$ respectively for $j \in [r_k]$ and $j \in [d_k] \setminus [r_k]$. And, $\hat{r}_k = r_k$ in probability for each $k \in [K]$.

Remark 2. Note that $\sqrt{d_k/(Td_{-k})}$ is dominated by both $T^\delta(d_k r_k)^{1/2}/d_{-k}^{1/2}$ (or $(d_k r_k/(Td_{-k}))^{1/2}$) and $(d_k/(Td_{-k}))^{1/4}$. Hence $b_T(\delta_2), b'_T(\delta_2) = o(1)$ necessarily means that $d_k/(Td_{-k}) = o(1)$ for each $k \in [K]$.

Consider $K = 3$, $d_1 = d_2 = d_3 = T$ and each r_k is finite. Suppose each factor loading matrix $\mathbf{A}_k$ has at least one pervasive factor (so $\delta_1 = 1$ in (3.10)), but with significantly weaker factors accompanied with $\delta_2 = 1/4$. Finally, let r_e be finite, and there are $O(T^{1/3})$ of the $\Psi_\ell^{(k)}$'s being non-zero with $\|\Psi_\ell^{(k)}\|_F$ finite, $\ell \in [d_{-k}]$. Then for each $k \in [K]$,

$$\lambda_j(\hat{\mathbf{R}}^{(k)}) = \begin{cases} \lambda_j(\mathbf{R}^{(k)})(1 + O_{a.s.}(T^{-1/4})), & j \in [r_k]; \\ \lambda_j(\mathbf{R}^{(k)}) + O_{a.s.}(T^{-1/2}), & j \in [d_k] \setminus [r_k]. \end{cases}$$

If $\delta_2 = 0$, then the first rate of $T^{-1/4}$ above will be replaced by $T^{-1/2}$. Clearly with Assumption (E3) in place, the rates of convergence for the eigenvalues of $\hat{\mathbf{R}}^{(k)}$ are better than those in Theorem 2, even when there are weak factors.

Moreover, if $K = 2$ for matrix-valued data, we still have almost sure convergence for $\lambda_j(\hat{\mathbf{R}}^{(k)})$ for each $k = 1, 2$, which is a major improvement from Theorem 2. For instance if $d_1 = d_2 = T$, $\delta_2 = 0$ and $\Psi^{(k)} = \mathbf{0}$, the almost sure rate of convergence in Theorem 3 is $b_T(0) = T^{-1/4}$ for each $k = 1, 2$. Whereas even if $d_1 = d_2 = T^{2/3}$, $\delta_2 = 0$ and $\Psi^{(k)} = \mathbf{0}$, the convergence is only in probability with rate $T^{-1/6}$ in Theorem 2.

5 Practical Implementation

In practice, it is not difficult for $\lambda_j(\hat{\mathbf{R}}^{(k)})$ with $j \in [d_k] \setminus [r_k]$ to be slightly larger than 1 because of small errors (with rate specified in Theorem 2 and 3). After all, we have $\lambda_j(\mathbf{R}^{(k)}) \leq 1$, and if the true value of $\lambda_j(\mathbf{R}^{(k)})$ is already very close to 1, then just a small error can tip $\lambda_j(\hat{\mathbf{R}}^{(k)})$ to be larger than 1, rendering $\hat{r}_k$ overestimating r_k .

To circumvent this practical problem, we can consider a small quantity η_T relating to the rates of

convergence $a_T(0)$ or $b_T(0)$ in Theorem 2 and 3 respectively, so that for each $k \in [K]$, we modify $\hat{r}_k$ from (3.13) to

$$\hat{r}_k = \{j : \lambda_j(\hat{\mathbf{R}}^{(k)}) > 1 + \eta_T, j \in [d_k]\}. \quad (5.16)$$

Since we do not know if Assumption (E3) is satisfied in practice, a more conservative and convenient quantity to consider is $\eta_T = \lambda\sqrt{d_k/T}$ from Theorem 2 for some constant $\lambda > 0$, when other rates from $a_T(0)$ are either dominated by $\sqrt{d_k/T}$ or unknown without further estimations. The larger the constant λ we set, the less likely that very weak factors can be detected. Through our extensive simulations, we find that setting $\lambda = 1$ strikes a good balance between being able to detect very weak factors and not overestimating r_k . In fact, note that in Fan et al. (2020), they use a threshold $1 + \sqrt{p/(n-1)}$ where p is the dimension of the observed data vectors, supported by in-depth theoretical analysis when $p/n \rightarrow c > 0$.

6 Empirical Studies

We present our simulation results in this section, which include comparisons of our proposed method TCorTh (using $\eta_T = \sqrt{d_k/T}$ in $\hat{r}_k$ defined in (5.16)) with the eigenvalue ratios (ER) and information criterion (IC) methods from Han et al. (2020). These two methods need estimated inputs, which can be from either the iTOPUP or iTIPUP iterative procedures described in Han et al. (2020). We iterate both to convergence, giving final rank estimators as described in their paper. We present two particular versions of ER and IC which used two particular penalty functions respectively, since other penalty functions suggested in Han et al. (2020) result in very similar performance. The IC method that used iTOPUP is abbreviated as iTOP-IC, while that used iTIPUP is abbreviated as iTIP-IC. Similarly there are two other methods, iTOP-ER and iTIP-ER. We use $\nu = 0$ and $c_0 = 0.1$ as suggested in Han et al. (2020) for tuning parameters of IC and ER respectively, while we fix $h_0 = 1$ as in Han et al. (2020) for both. For more details on these methods, please refer to Han et al. (2020).

For generating our data, we use model (3.4), with elements in μ being i.i.d. standard normal in each repetition of experiment. Each factor loading matrix $\mathbf{A}_k$ is generated independently with $\mathbf{A}_k = \mathbf{U}_k \mathbf{R}_k$, where the elements in $\mathbf{U}_k \in \mathbb{R}^{d_k \times r_k}$ are i.i.d. $U(-2, 2)$, and $\mathbf{R}_k \in \mathbb{R}^{r_k}$ is diagonal with the j th diagonal element being $d_k^{-\zeta_{k,j}}$, $0 \leq \zeta_{k,j} \leq 0.5$. Pervasive (strong) factors have $\zeta_{k,j} = 0$, while weak factors have $0 < \zeta_{k,j} \leq 0.5$. See Lam et al. (2011) or Han et al. (2020) for the definition of the strength of factors.

The elements in $\mathcal{F}_t$ are independent standardized AR(5) with AR coefficients 0.7, 0.3, -0.4, 0.2 and -0.1. The corresponding innovation process is standardized i.i.d. t_5 . Same for the elements in $\mathbf{e}_t^{(k)}$ and $\boldsymbol{\epsilon}_{t,\ell}^{(k)}$ in (3.6), but their AR coefficients are (-0.7, -0.3, -0.4, 0.2, 0.1) and (0.8, 0.4, -0.4, 0.2, -0.1) respectively. The

matrix $\Psi^{(1)} \in \mathbb{R}^{d_1 \times d_1}$ is generated with i.i.d. standard normal entries, but has an independent probability of 0.7 being set exactly to 0. Each matrix $\Sigma_{\epsilon, \ell}^{(1)}$ has randomly generated eigenvectors with eigenvalues on $U(1, 3)$. In some settings where Assumption (E3) is satisfied, the $w_\ell^{(1)}$'s are i.i.d. $U(1, 2)$, and the $\sigma_j^{(1)}$'s are i.i.d. $U(0.1, 2.1)$. Each experiment is repeated 1000 times.

We consider the following simulation settings:

- (I) Set $K = 2$, $r_1 = 3, r_2 = 2$, $d_1 = d_2 = 40$, $T = 200$. All strong factors with $\zeta_{k,j} = 0$ for all k, j . Assumption (E3) is satisfied.
- (II) Same as I, but $d_1 = 140$ and Assumption (E3) is not satisfied.
- (III) Same as II, except that $\zeta_{k,r_k} = 0.2$. All other $\zeta'_{k,j}$ s remain 0.
- (IV) Set $K = 2$, $r_1 = 3, r_2 = 2$, $d_1 = 30, d_2 = 20$, $T = 100$. Also, each $\zeta_{k,j} = 0$ except $\zeta_{k,r_k} = 0.2$. Assumption (E3) is satisfied.
- (V) Same as (IV), except that $\zeta_{k,r_k} = 0.3$.
- (VI) Same as (V), except $\zeta_{k,r_k-1} = 0.15$. We try combinations of $T = 100, 200, 400$ and $(d_1, d_2) = (30, 20), (60, 40)$.
- (VII) Set $K = 3$, $r_1 = r_2 = r_3 = 3$, $d_1 = d_2 = d_3 = 40$, $T = 200$. Also, $\zeta_{k,1} = 0, \zeta_{k,2} = 0.1$ and $\zeta_{k,3} = 0.2$. Assumption (E3) is satisfied.
- (VIII) Same as (VII), except that $T = 400$, $\zeta_{k,2} = 0.15$ and $\zeta_{k,3} = 0.3$.

Setting (I) to (VI) have $K = 2$ while setting (VII) and (VIII) have $K = 3$. We would want to test larger K , but the simulation of such a huge tensor proves to be too time consuming. Setting (I) and (II) have all strong factors while Assumption (E3) is in place only for (II). All other settings have weak factors, and we explore the effects of mixing in such weak factors with various strength under these remaining settings. In particular, setting (VI) has 3 layers of strength of factors, from strong to very weak, with one factor loading matrix comprised entirely of weak factors. We investigate the performance of different methods under different combinations of T and $d_1, \dots, d_K$ in such a setting. The choice of t_5 as the innovations in all time series processes is to present more challenges to the methods to be compared. Gaussian innovations are used also, but their results for iTIP-ER and TCorTh, which are the two methods to be compared in the majority of the section, do not differ much from their t_5 counterparts. Hence we omit their presentations.

The first thing we can learn from Table 1 is that the IC methods are not reliable, which is to be expected since our generated random variables are of time series nature, and IC is not created to deal with

Setting (I)	Rank estimator		Correct	Run
Method	$\hat{r}_1$	$\hat{r}_2$	Proportion	Time(s)
iTOP-IC	15 ₍₀₎	15 ₍₀₎	0	11.4 _(.2)
iTIP-IC	6.7 _(1.3)	6.5 _(1.7)	0	.13 _(.01)
iTOP-ER	3 ₍₀₎	2 ₍₀₎	1	11.4 _(.2)
iTIP-ER	3 ₍₀₎	2 ₍₀₎	1	.13 _(.01)
TCorTh	3 ₍₀₎	2 ₍₀₎	1	.0034 _(.0005)

Table 1: Results for setting (I). The columns $\hat{r}_1$ and $\hat{r}_2$ report the mean and SD (subscripted, in bracket) of the estimates $\hat{r}_1$ and $\hat{r}_2$ respectively under different methods. Correct proportion counts the proportion of $(\hat{r}_1, \hat{r}_2) = (3, 2)$. Run time is the mean and SD of the run time in seconds.

this. Hence in the remainder of Section 6, we do not report results on IC which are completely inaccurate anyway. However, ER methods perform perfectly, as with our method TCorTh which is computationally much less demanding. At the same time, the iTOPUP methods require much longer run time for each simulation to produce results. The majority of time is used to compute the initial iTOPUP estimators for the factor loading matrices, which requires the handling of large 5 dimensional tensors. In fact, in setting (II) where $d_1 = 140$, the tensor dimension that iTOPUP needs to handle is $140 \times 40 \times 140 \times 40 \times 1$ if $h_0 = 1$, which is already very large, and an average run time exceeds 100 seconds. Since we find that iTOP-ER is in fact slightly worse than iTIP-ER in all settings, we decide to report only iTIP-ER in the remainder of this section.

	iTIP-ER			
Setting	$\hat{r}_1$	$\hat{r}_2$	Correct Proportion	Run Time(s)
(II)	3 ₍₀₎	2 ₍₀₎	1	.4 _(.02)
(III)	2.84 _(.36)	2 ₍₀₎	.844	.5 _(.1)
	TCorTh			
(II)	3 ₍₀₎	2 ₍₀₎	1	.01 _(.001)
(III)	3 ₍₀₎	2 ₍₀₎	1	.01 _(.002)

Table 2: Results for setting (II) and (III). Refer to Table 1 for explanations.

Comparing the run time in Table 1 and Table 2, it is clear that the more than three fold increase in d_1 increases all run times around 3 times as well (setting (II)). From Table 2, with weak factors mixed in (setting (III)), there is also a slight increase in all run times. TCorTh still works perfectly in both settings, while iTIP-ER underestimates r_1 in around 16% of the cases.

From Table 3, it is clear that the adverse effects on the performance on all rank estimators are stronger when factors are weaker. From setting (IV) with $\zeta_{k,r_k} = 0.2$ to (V) with $\zeta_{k,r_k} = 0.3$, and to setting (VI) with even more mixed weak factors, the proportion of correct rank estimation comes down from 93.4% to 43.6% and to only 25% respectively for TCorTh. For iTIP-ER, the decrease is more dramatic, from 56.8% to 5.6% and to only 2%. Not surprisingly, the majority of cases where the rank estimators are wrong are

		iTIP-ER		TCorTh		iTIP-ER	TCorTh	iTIP-ER	TCorTh
Setting		$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	Correct Proportion		Run Time(s)	
(IV)		2.57 _(.57)	1.92 _(.28)	2.94 _(.24)	2.00 _(.06)	.568		.06 _(.008)	
(V)		2.07 _(.44)	1.54 _(.50)	2.47 _(.51)	1.94 _(.23)	.056		.06 _(.009)	
(VI)		iTIP-ER		TCorTh		iTIP-ER	TCorTh	iTIP-ER	TCorTh
T	(d_1, d_2)	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	Correct Proportion		Run Time(s)	
100	(30,20)	1.72 _(1.43)	1.79 _(1.09)	2.26 _(.45)	1.96 _(.21)	.02		.07 _(.01)	
	(60,40)	1.50 _(0.68)	1.77 _(.47)	2.13 _(.33)	2.00 _(.08)	.014		.2 _(.02)	
200	(30,20)	1.51 _(.59)	1.86 _(.35)	2.58 _(.50)	1.99 _(.08)	.034		.1 _(.02)	
	(60,40)	1.61 _(.59)	1.97 _(.18)	2.42 _(.49)	2 ₍₀₎	.056		.2 _(.03)	
400	(30,20)	1.67 _(.67)	1.95 _(.23)	2.90 _(.39)	2 ₍₀₎	.108		.2 _(.03)	
	(60,40)	1.79 _(.73)	2.00 _(.04)	2.92 _(.31)	2 ₍₀₎	.186		.4 _(.04)	

Table 3: Results for setting (IV), (V) and (VI). True $(r_1, r_2) = (3, 2)$. Refer to Table 1 for explanations.

underestimation rather than overestimation in the face of the inability to detect the weakest factors. In fact only several of the cases are overestimation for TCorTh, as somehow reflected in the means of $\hat{r}_1$ and $\hat{r}_2$ and the overall small SD of $\hat{r}_1$ and $\hat{r}_2$ in the settings involved. Interestingly though, for iTIP-ER, under setting (VI) when $T = 100$ and $(d_1, d_2) = (30, 20)$, there are around 40 cases of overestimation for r_1 or r_2 , as seen in the relatively large SD of $\hat{r}_1$ and $\hat{r}_2$ in this particular case.

Under setting (VI), we see that as T increases, the performance of all estimators are better in general while they get worse when dimension increases, except when $T = 400$, where correct proportion of estimation actually increases for both methods when dimensions increase. The latter can be down to the fact that when dimensions increase, so does the r_k -th largest eigenvalue for $\mathbf{R}^{(k)}$. The cases for smaller T just may not be able to rip such benefits because of the larger errors from estimations as dimensions increase. Overall, we can see that TCorTh has superior performance in terms of accuracy and run time over iTIP-ER on average under different scenarios.

iTIP-ER					
Setting	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_3$	Correct Proportion	Run Time(s)
(VII)	3 ₍₀₎	3 ₍₀₎	3 ₍₀₎	1	1.5 _(.2)
(VIII)	3 ₍₀₎	3 ₍₀₎	3 ₍₀₎	1	3.8 _(.8)
TCorTh					
(VII)	3 ₍₀₎	3 ₍₀₎	3 ₍₀₎	1	.1 _(.01)
(VIII)	3.00 _(.06)	3.00 _(.06)	3.00 _(.06)	.988	.3 _(.02)

Table 4: Results for setting (VII) and (VIII). True $(r_1, r_2, r_3) = (3, 3, 3)$. Refer to Table 1 for explanations.

Finally, Table 4 summarized the results when $K = 3$. It is clear that both methods are still doing well even in the presence of a mix of weak factors, but with iTIP-ER having a slight edge when some factors are very weak in setting (VIII).

6.1 Real data analysis

We analyze a set of Fama-French portfolio returns data formed on size and operating profitability. Stocks are categorized into 10 different sizes (market equity, using NYSE market equity deciles) and 10 different operating profitability (OP) levels (using NYSE OP deciles. OP is annual revenues minus cost of goods sold, interest expense, and selling, general, and administrative expenses divided by book equity for the last fiscal year end). These levels, and hence the stocks in each category, are allocated at the end of June each year. Moreover, the stocks in each of the 10×10 categories form exactly two portfolios, one being value weighted, and the other of equal weight. Hence, there are two sets of 10×10 portfolios with their time series of returns observed. We use monthly data from July 1973 to June 2021, so that $T = 576$, and each data tensor we have thus has size $10 \times 10 \times 576$. For more details, please visit

https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_100_port_szme_op.html

Since the market factor is certainly pervasive in financial returns, we use the following CAPM to remove its effects and facilitate detection of potentially weaker factors:

$$\mathbf{y}_t = \bar{\mathbf{y}} + \boldsymbol{\beta}(x_t - \bar{x}) + \mathbf{e}_t,$$

where $\mathbf{y}_t \in \mathbb{R}^{100}$ contains the returns of the 100 portfolios at time t , x_t is the return for the NYSE composite index at time t , and $\boldsymbol{\beta} = (\beta_1, \dots, \beta_{100})^\top$ is the vector of β 's for the 100 portfolios. Least squares estimation leads us to

$$\hat{\boldsymbol{\beta}} = \frac{\sum_{t=1}^T (x_t - \bar{x})(\mathbf{y}_t - \bar{\mathbf{y}})}{\sum_{t=1}^T (x_t - \bar{x})^2}.$$

Hence the data we analyze is $\{\mathbf{y}_t - \bar{\mathbf{y}} - \hat{\boldsymbol{\beta}}(x_t - \bar{x})\}_{t=1, \dots, 576}$, with each observed vector reshaped into a 10×10 tensor.

	Value Weighted								Equal Weight							
	$h_0 = 1$		$h_0 = 2$		$h_0 = 3$		$h_0 = 4$		$h_0 = 1$		$h_0 = 2$		$h_0 = 3$		$h_0 = 4$	
Method	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$	$\hat{r}_1$	$\hat{r}_2$
iTOP-IC	4	2	4	5	6	6	6	6	4	3	5	6	6	6	6	6
iTIP-IC	2	2	3	3	4	3	5	3	2	3	3	4	4	4	5	4
iTOP-ER	3	1	4	1	1	1	1	1	3	1	1	1	1	1	1	1
iTIP-ER	5	2	4	1	1	1	1	1	4	3	1	1	1	1	1	1
TCorTh	1	2	-	-	-	-	-	-	1	2	-	-	-	-	-	-

Table 5: Results for Value-weighted and equal weight portfolios.

The results are presented in Table 5. While each h_0 produces different results for the IC and ER methods, and the magnitudes of $\hat{r}_k$ go in opposite directions in general for IC and ER respectively when h_0 increases, it would seem that $(\hat{r}_1, \hat{r}_2) = (1, 1)$ is more sensible from all of the possibilities.

Our method TCorTH gives $(\hat{r}_1, \hat{r}_2) = (1, 2)$. Looking at the eigenvalues of the total mode-1 correlation matrices from Figure 1, both value weighted and equal weight portfolios are having their second eigenvalues below 1, and hence the $1 + \sqrt{d_1/T}$ threshold. Hence it is quite clear cut that there should only be 1 strong factor, and hence $\hat{r}_1 = 1$ for both cases. For the total mode-2 correlation matrices, we see that both the second eigenvalues are above the threshold (for equal weight case, only just), with the third eigenvalues well below 1 and the threshold. Hence $\hat{r}_2 = 2$ seems to be a good choice, although it could be marginal for the equal weight case.

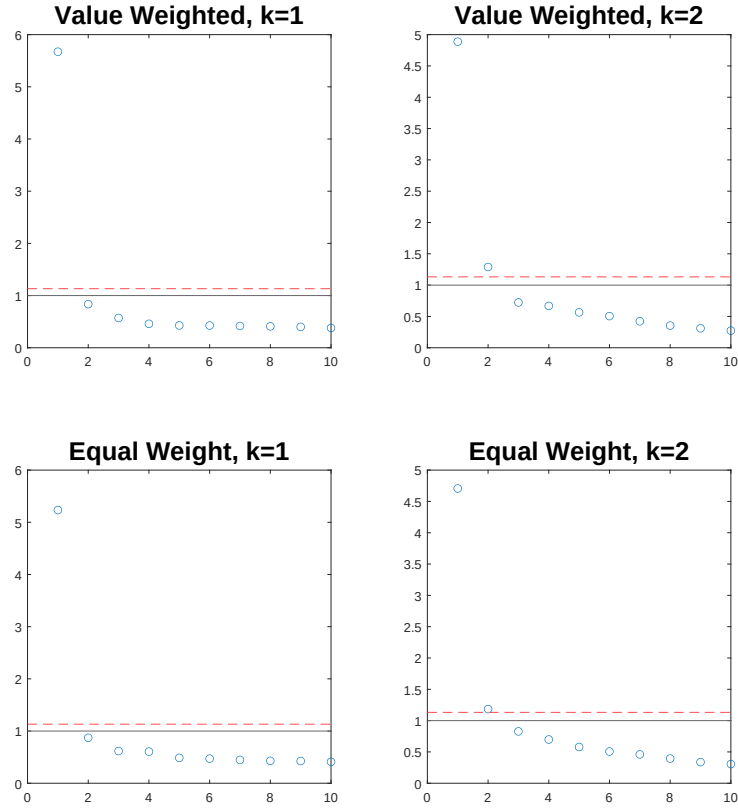


Figure 1: Plot of eigenvalues for the total mode-1 and mode-2 correlation matrices for the value weighted and equal weight portfolio data. The horizontal dashed line in each plot represents $1 + \sqrt{10/576} = 1.13$. *Upper row:* value weighted. *Lower row:* equal weight. *Left column:* $k = 1$. *Right column:* $k = 2$.

7 Conclusion

We develop a practical rank estimator for tensor time series factor model with low computational burden relative to other state-of-the-art alternatives. This makes our method aligns better with larger dimensional tensors with a moderate K . While the theoretical development is based on either d_k/T or $d_k/(Td_{-k}) = o(1)$, many financial and economic applications satisfy both conditions easily. That said, further development assuming $d_k/T \rightarrow c > 0$ with a more general error structure will certainly extend the scope of our method.

Since our method is not coupled with estimation of factor loading matrices, a method of projection like the one used in iTOPUP or iTIPUP in Han et al. (2020) is not possible yet for our method. In the future, we aim to develop estimation procedure for a time series tensor factor model under general econometric assumptions so as to be able to further complete such an important tool in finance, economics and other science disciplines. At the same time, this will give us inputs for developing an iterative procedure for potentially more accurate estimation of the rank of the core tensor in a tensor factor model.

8 Proofs

We present an important lemma below first, which is a simplified version of Theorem 2.8 of Wang and Paul (2014). This lemma is central to the proofs of Theorem 2 and 3.

Lemma 1. *Let $B_n \in \mathbb{R}^{n \times n}$ be positive semi-definite with $\|B_n\|$ bounded above, $n^{-1}\text{tr}(B_n) \rightarrow b_1$ and $n^{-1}\text{tr}(B_n^2) \rightarrow b_2$ as $n \rightarrow \infty$. Let $Y_n = X_n B_n^{1/2}$, where $X_n \in \mathbb{R}^{p \times n}$ has i.i.d. sub-Gaussian entries (unless $p = o(n^{1/3})$, only finiteness of EX_{11}^4 assumed then) with $EX_{11} = 0, EX_{11}^2 = 1$. Let $S_n = n^{-1}Y_n Y_n^\top$. Then as $p, n \rightarrow \infty$ such that $p/n \rightarrow 0$, the empirical distribution of $\{\sqrt{n/p}(\lambda_j(S_n) - n^{-1}\text{tr}(B_n))\}_{j=1}^p$ converges almost surely to a nonrandom probability distribution $F(x) = F_{sc}(b_2^{-1/2}x)$, $x \in \mathbb{R}$, where F_{sc} is the semi-circle law.*

Remark 3. Since the semi-circle law has finite support with length directly proportional to $b_2^{1/2}$, it is clear that (by considering subsequences that converge) even if we only have $n^{-1}\text{tr}(B_n) \leq b_1$, $n^{-1}\text{tr}(B_n^2) \leq b_2$ (both implied by the boundedness of $\|B_n\|$), we can still conclude that

$$\{\sqrt{n/p}(\lambda_j(S_n) - n^{-1}\text{tr}(B_n))\}_{j=1}^p = O_{a.s.}(1). \quad (8.17)$$

Proof of Theorem 1. From (3.11), for each $k \in [K]$,

$$\mathbf{R}^{(k)} = \mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top} + \mathbf{Q}_2^{(k)\top} \mathbf{Q}_2^{(k)\top} + \mathbf{Q}_3^{(k)} \mathbf{Q}_3^{(k)\top}, \text{ where}$$

$$\mathbf{Q}_1^{(k)} = \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \|\mathbf{A}_{-k}\|_F \mathbf{A}_k, \quad \mathbf{Q}_2^{(k)} = \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \boldsymbol{\Sigma}_\epsilon^{(k)}, \quad \mathbf{Q}_3^{(k)} = \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \boldsymbol{\Psi}^{(k)}.$$

Since $\mathbf{R}^{(k)}$ has a diagonal of ones by construction, we have

$$d_k = \text{tr}(\mathbf{R}^{(k)}) \geq \text{tr}(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}) = \|\mathbf{A}_{-k}\|_F^2 \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|_F^2 \geq \|\mathbf{A}_{-k}\|_F^2 \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|^2 \geq C d_k^{\delta_1}, \quad (8.18)$$

where the last inequality is by Assumption (M1). The above shows that the assumption itself is a reasonable one, and that $\delta_1 \in [0, 1]$. Now using Assumption (M1) again, consider

$$\frac{\lambda_1(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top})}{\lambda_{r_k}(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top})} = \frac{\lambda_1(\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k)}{\lambda_{r_k}(\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k)} = \|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)}) \mathbf{A}_k\| \cdot \|\{\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)}) \mathbf{A}_k\}^{-1}\| = O(d_k^{\delta_2}),$$

so that combined with (8.18), for some generic constant $C' > 0$,

$$C d_k^{\delta_1} \leq \text{tr}(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}) = \sum_{j=1}^{r_k} \lambda_j(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}) \leq r_k \lambda_1(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}) \leq C' r_k d_k^{\delta_2} \lambda_{r_k}(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}).$$

The above means that, as T, d_k grows large together for all $k \in [K]$,

$$\lambda_{r_k}(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}) \geq C/C' \cdot d_k^{\delta_1 - \delta_2} r_k^{-1} > 1$$

for large enough T, d_k for $k \in [K]$, since $r_k = o(d_k^{\delta_1 - \delta_2})$ by Assumption (M1). This completes the proof of $\lambda_j(\mathbf{R}^{(k)}) > 1$ for $j \in [r_k]$ for each $k \in [K]$.

Finally, consider for $j = r_k + 1, \dots, d_k$,

$$\lambda_j(\mathbf{R}^{(k)}) \leq \lambda_j(\mathbf{Q}_1^{(k)} \mathbf{Q}_1^{(k)\top}) + \lambda_1(\mathbf{Q}_2^{(k)} \mathbf{Q}_2^{(k)\top} + \mathbf{Q}_3^{(k)} \mathbf{Q}_3^{(k)\top}) = 0 + \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} (\boldsymbol{\Sigma}_\epsilon^{(k)} + \boldsymbol{\Psi}^{(k)} \boldsymbol{\Psi}^{(k)\top})\| \leq 1$$

by Assumption (M1). This completes the proof of the Theorem. $\square$

Before the proofs of our main theorems, define

$$\mathbf{S}^{(k)} := \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \widehat{\boldsymbol{\Sigma}}^{(k)} \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2}. \quad (8.19)$$

From (8.19), it is not difficult to see that

$$\widehat{\mathbf{R}}^{(k)} = \text{diag}(\mathbf{S}^{(k)})^{-1/2} \mathbf{S}^{(k)} \text{diag}(\mathbf{S}^{(k)})^{-1/2}. \quad (8.20)$$

Using (3.5) and (3.6), we can write the j th column of $\text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}})$ as

$$\begin{aligned} (\text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}}))_j &= \mathbf{A}_k \text{mat}_k(\mathcal{F}_t - \bar{\mathcal{F}})(\mathbf{A}_{-k}^T)_j + (\text{mat}_k(\mathcal{E}_t - \bar{\mathcal{E}}))_j \\ &= \mathbf{A}_k(\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k})_j^T) \text{vec}(\text{mat}_k(\mathcal{F}_t - \bar{\mathcal{F}})^T) + (\boldsymbol{\Sigma}_{\epsilon,j}^{(k)})^{1/2}(\boldsymbol{\epsilon}_{t,j}^{(k)} - \bar{\boldsymbol{\epsilon}}_{t,j}^{(k)}) + \boldsymbol{\Psi}_j^{(k)}(\mathbf{e}_t^{(k)} - \bar{\mathbf{e}}^{(k)}) \\ &=: \mathbf{A}_k(\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k})_j^T)(\mathbf{f}_t^{(k)} - \bar{\mathbf{f}}^{(k)}) + (\boldsymbol{\Sigma}_{\epsilon,j}^{(k)})^{1/2}(\boldsymbol{\epsilon}_{t,j}^{(k)} - \bar{\boldsymbol{\epsilon}}_{t,j}^{(k)}) + \boldsymbol{\Psi}_j^{(k)}(\mathbf{e}_t^{(k)} - \bar{\mathbf{e}}^{(k)}), \end{aligned}$$

where we use $(\cdot)_j$ to denote the j th column of a matrix. With the above, defining $\mathbf{M}_T := \mathbf{I}_T - T^{-1} \mathbf{1}_T \mathbf{1}_T^T$, $\mathbf{X}_j^{(k)} := (\mathbf{x}_{1,j}^{(k)}, \dots, \mathbf{x}_{T,j}^{(k)})$ with $\mathbf{x}_{t,j}^{(k)} := (\mathbf{f}_t^{(k)T}, \boldsymbol{\epsilon}_{t,j}^{(k)T}, \mathbf{e}_t^{(k)T})^T$, we can write

$$\begin{aligned} \mathbf{S}^{(k)} &= T^{-1} \sum_{t=1}^T \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}}) \text{mat}_k(\mathcal{X}_t - \bar{\mathcal{X}})^T \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \\ &= T^{-1} \sum_{j=1}^{d_k} (\mathbf{Q}_{1j}^{(k)}, \mathbf{Q}_{2j}^{(k)}, \mathbf{Q}_{3j}^{(k)}) \mathbf{X}_j^{(k)} \mathbf{M}_T \mathbf{X}_j^{(k)T} (\mathbf{Q}_{1j}^{(k)}, \mathbf{Q}_{2j}^{(k)}, \mathbf{Q}_{3j}^{(k)})^T, \text{ where} \\ (\mathbf{Q}_{1j}^{(k)}, \mathbf{Q}_{2j}^{(k)}, \mathbf{Q}_{3j}^{(k)}) &:= \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} (\mathbf{A}_k(\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k})_j^T), (\boldsymbol{\Sigma}_{\epsilon,j}^{(k)})^{1/2}, \boldsymbol{\Psi}_j^{(k)}). \end{aligned}$$

Now define $\tilde{\mathbf{F}}^{(k)} := (\mathbf{f}_1^{(k)}, \dots, \mathbf{f}_T^{(k)})$, $\tilde{\boldsymbol{\Xi}}_j^{(k)} := (\boldsymbol{\epsilon}_{1,j}^{(k)}, \dots, \boldsymbol{\epsilon}_{T,j}^{(k)})$ and $\tilde{\mathbf{E}}^{(k)} := (\mathbf{e}_1^{(k)}, \dots, \mathbf{e}_T^{(k)})$. Then using the decomposition of $\mathbf{S}^{(k)}$ above, we can write $\mathbf{S}^{(k)} = \sum_{i,j=1}^3 \tilde{\mathbf{S}}_{ij}^{(k)}$, where $\tilde{\mathbf{S}}_{ij}^{(k)} = \tilde{\mathbf{S}}_{ji}^{(k)T}$ for $i \neq j$, and

$$\begin{aligned} \tilde{\mathbf{S}}_{11}^{(k)} &:= T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \tilde{\mathbf{F}}^{(k)} \mathbf{M}_T \tilde{\mathbf{F}}^{(k)T} \mathbf{Q}_{1j}^{(k)T}, \quad \tilde{\mathbf{S}}_{12}^{(k)} := T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \tilde{\mathbf{F}}^{(k)} \mathbf{M}_T \tilde{\boldsymbol{\Xi}}_j^{(k)T} \mathbf{Q}_{2j}^{(k)T}, \\ \tilde{\mathbf{S}}_{13}^{(k)} &:= T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \tilde{\mathbf{F}}^{(k)} \mathbf{M}_T \tilde{\mathbf{E}}^{(k)T} \mathbf{Q}_{3j}^{(k)T}, \quad \tilde{\mathbf{S}}_{22}^{(k)} := T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{2j}^{(k)} \tilde{\boldsymbol{\Xi}}_j^{(k)} \mathbf{M}_T \tilde{\boldsymbol{\Xi}}_j^{(k)T} \mathbf{Q}_{2j}^{(k)T}, \\ \tilde{\mathbf{S}}_{23}^{(k)} &:= T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{2j}^{(k)} \tilde{\boldsymbol{\Xi}}_j^{(k)} \mathbf{M}_T \tilde{\mathbf{E}}^{(k)T} \mathbf{Q}_{3j}^{(k)T}, \quad \tilde{\mathbf{S}}_{33}^{(k)} := T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{3j}^{(k)} \tilde{\mathbf{E}}^{(k)} \mathbf{M}_T \tilde{\mathbf{E}}^{(k)T} \mathbf{Q}_{3j}^{(k)T}. \end{aligned} \quad (8.21)$$

With the notations in (E2) and (F1), define $\mathbf{z}_{e,t}^{(k)} := (z_{e,t,1}^{(k)}, \dots, z_{e,t,r_e}^{(k)})^T$, $\mathbf{z}_{\epsilon,t,j}^{(k)} := (z_{\epsilon,t,j,1}^{(k)}, \dots, z_{\epsilon,t,j,d_k}^{(k)})^T$ and

$\mathbf{z}_{f,t}^{(k)} := (z_{f,t,1,1}, \dots, z_{f,t,1,r_k}, \dots, z_{f,t,r_k,1}, \dots, z_{f,t,r_k,r_k})^T$. We further split

$$\begin{aligned}\tilde{\mathbf{F}}^{(k)} &= \mathbf{F}^{(k)} + \check{\mathbf{F}}^{(k)}, \quad \tilde{\Xi}_j^{(k)} = \Xi_j^{(k)} + \check{\Xi}_j^{(k)}, \quad \tilde{\mathbf{E}}^{(k)} = \mathbf{E}^{(k)} + \check{\mathbf{E}}^{(k)}, \quad \text{where} \\ \mathbf{F}^{(k)} &:= \left(\sum_{q=1}^{mT} a_{f,q} \mathbf{z}_{f,1-q}^{(k)}, \dots, \sum_{q=1}^{mT} a_{f,q} \mathbf{z}_{f,T-q}^{(k)} \right) = (\mathbf{z}_{f,1-mT}^{(k)}, \dots, \mathbf{z}_{f,T}^{(k)}) \mathbf{A}_{f,T} =: \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T}, \\ \Xi_j^{(k)} &:= \left(\sum_{q=1}^{mT} a_{\epsilon,q} \mathbf{z}_{\epsilon,1-q,j}^{(k)}, \dots, \sum_{q=1}^{mT} a_{\epsilon,q} \mathbf{z}_{\epsilon,T-q,j}^{(k)} \right) = (\mathbf{z}_{\epsilon,1-mT,j}^{(k)}, \dots, \mathbf{z}_{\epsilon,T,j}^{(k)}) \mathbf{A}_{\epsilon,T} =: \mathbf{Z}_{\epsilon,j}^{(k)} \mathbf{A}_{\epsilon,T}, \\ \tilde{\mathbf{E}}^{(k)} &:= \left(\sum_{q=1}^{mT} a_{e,q} \mathbf{z}_{e,1-q}^{(k)}, \dots, \sum_{q=1}^{mT} a_{e,q} \mathbf{z}_{e,T-q}^{(k)} \right) = (\mathbf{z}_{e,1-mT}^{(k)}, \dots, \mathbf{z}_{e,T}^{(k)}) \mathbf{A}_{e,T} =: \mathbf{Z}_e^{(k)} \mathbf{A}_{e,T},\end{aligned}$$

with $\mathbf{A}_{f,T}$, $\mathbf{A}_{e,T}$ and $\mathbf{A}_{\epsilon,T}$ defined in Assumption (R2), and $\check{\mathbf{F}}^{(k)}$, $\check{\Xi}_j^{(k)}$ and $\check{\mathbf{E}}^{(k)}$ the remainders of the truncations. Then we define

$$\begin{aligned}\tilde{\mathbf{S}}_{ij}^{(k)} &= \mathbf{S}_{ij}^{(k)} + \check{\mathbf{S}}_{ij}^{(k)}, \quad \text{where} \\ \mathbf{S}_{11}^{(k)} &:= T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{F}^{(k)} \mathbf{M}_T \mathbf{F}^{(k)\top} \mathbf{Q}_{1j}^{(k)\top},\end{aligned}\tag{8.22}$$

and similarly for other $\mathbf{S}_{ij}^{(k)}$'s. We first prove a lemma on how small each remainder term $\check{\mathbf{S}}_{ij}^{(k)}$ is.

Lemma 2. *Let Assumptions (E1), (E2), (F1), (M1) and (R2) hold. Then for each $k \in [K]$,*

$$\sum_{i,j=1}^3 \|\check{\mathbf{S}}_{ij}^{(k)}\| = O_{a.s.}(rT^{\delta-1/2}d_k^{-1}),$$

where $0 < \delta < 1/2$ is some small constant.

Proof of Lemma 2. Firstly, by Assumption (E2), (F1), (M1) and (R2), we have

$$\begin{aligned}\sum_{j=1}^{d_k} \|\mathbf{Q}_{1j}^{(k)}\|^2 &= \|\mathbf{A}_{\cdot k}\|_F^2 \cdot \|\text{diag}(\mathbf{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|^2 < d_k, \quad \sum_{j=1}^{d_k} \|\mathbf{Q}_{2j}^{(k)}\|^2 \leq \sum_{j=1}^{d_k} \|\text{diag}(\mathbf{\Sigma}^{(k)})^{-1} \mathbf{\Sigma}_{\epsilon,j}^{(k)}\| < \infty, \\ \sum_{j=1}^{d_k} \|\mathbf{Q}_{3j}^{(k)}\|^2 &\leq \sum_{j=1}^{d_k} \|\text{diag}(\mathbf{\Sigma}^{(k)})^{-1} \mathbf{\Psi}_j^{(k)} \mathbf{\Psi}_j^{(k)\top}\| < \infty, \\ E\|T^{-1/2} \mathbf{F}^{(k)}\|_F^2 &= T^{-1} E \text{tr}(\mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} \mathbf{Z}_{f,T}^{(k)} \mathbf{A}_{f,T}) = rT^{-1} \text{tr}(\mathbf{A}_{f,T}^\top \mathbf{A}_{f,T}) = r(1 - o(T^{-3}d^{-2})), \\ E\|T^{-1/2} \Xi_j^{(k)}\|_F^2 &= d_k(1 - o(T^{-3}d^{-2})), \quad E\|T^{-1/2} \mathbf{E}^{(k)}\|_F^2 = r_e(1 - o(T^{-3}d^{-2})), \\ E\|T^{-1/2} \check{\mathbf{F}}^{(k)}\|_F^2 &= o(rT^{-3}d^{-2}), \quad E\|T^{-1/2} \check{\Xi}_j^{(k)}\|_F^2 = o(d_k T^{-3}d^{-2}), \quad E\|T^{-1/2} \check{\mathbf{E}}^{(k)}\|_F^2 = o(r_e T^{-3}d^{-2}).\end{aligned}\tag{8.23}$$

We can write $\check{\mathbf{S}}_{11}^{(k)} = I_1 + I_1^T + I_2$, where

$$I_1 = T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \check{\mathbf{F}}^{(k)} \mathbf{M}_T \mathbf{F}^{(k)T} \mathbf{Q}_{1j}^{(k)T}, \quad I_2 = T^{-1} \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \check{\mathbf{F}}^{(k)} \mathbf{M}_T \check{\mathbf{F}}^{(k)T} \mathbf{Q}_{1j}^{(k)T}.$$

Let $a = rT^{\delta-1/2}d_k^{-1}$ for some small constant $0 < \delta < 1/2$. Using the Markov's inequality and (8.23),

$$\begin{aligned} P(\|I_1\| > a) &\leq \sum_{j=1}^{d_k} \|\mathbf{Q}_{1j}^{(k)}\|^2 \cdot E\|T^{-1/2}\check{\mathbf{F}}^{(k)}\| \|T^{-1/2}\mathbf{F}^{(k)}\|/a \\ &\leq d_k \cdot E^{1/2}\|T^{-1/2}\check{\mathbf{F}}^{(k)}\|_F^2 \cdot E^{1/2}\|T^{-1/2}\mathbf{F}^{(k)}\|_F^2/a \\ &= d_k \cdot \{o(rT^{-3}d^{-2})\}^{1/2} \cdot \{r(1 - o(T^{-3}d^{-2}))\}^{1/2}/a = O(rT^{-3/2}d_k^{-1}a^{-1}) = O(T^{-1-\delta}), \end{aligned}$$

Hence the Borel-Cantelli's lemma implies that $\|I_1\| = O_{a.s.}(rT^{\delta-1/2}d_k^{-1})$. Similarly (details omitted), $\|I_2\| = O_{a.s.}(rT^{\delta-2}d^{-1}d_k^{-1})$, so that $\|\check{\mathbf{S}}_{11}^{(k)}\| = O_{a.s.}(rT^{\delta-1/2}d_k^{-1})$.

Similar to the above arguments, using (8.23), we can show that

$$\begin{aligned} \|\check{\mathbf{S}}_{12}^{(k)}\| &= O_{a.s.}(r^{1/2}T^{\delta-1/2}d_k^{-1}), \quad \|\check{\mathbf{S}}_{13}^{(k)}\| = O_{a.s.}((rr_e/d_k)^{1/2}T^{\delta-1/2}d_k^{-1}), \quad \|\check{\mathbf{S}}_{22}^{(k)}\| = O_{a.s.}(T^{\delta-1/2}d_k^{-1}), \\ \|\check{\mathbf{S}}_{23}^{(k)}\| &= O_{a.s.}((r_e/d_k)^{1/2}T^{\delta-1/2}d_k^{-1}), \quad \|\check{\mathbf{S}}_{33}^{(k)}\| = O_{a.s.}((r_e/d_k)T^{\delta-1/2}d_k^{-1}). \end{aligned}$$

Since $r_e = o(d_k)$ by Assumption (M1), we can conclude that $\|\check{\mathbf{S}}_{11}^{(k)}\|$ dominates all other terms, and the proof completes. $\square$

Lemma 3. *Let Assumption (E1), (E2), (F1), (M1), (R1) and (R2) hold. Then for $j \in [r_k]$ and $k \in [K]$, and a small constant $\delta > 0$, we have*

$$\begin{aligned} \frac{\|\mathbf{S}_{11}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)T}\|}{\lambda_j(\mathbf{R}^{(k)})} &= O_{a.s.}\left(d_k^{\delta_2} \sqrt{\frac{r}{T}}\right), \quad \|\mathbf{S}_{22}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{2j}^{(k)} \mathbf{Q}_{2j}^{(k)T}\| = O_{a.s.}\left(\sqrt{\frac{d_k}{T}}\right), \\ \|\mathbf{S}_{33}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{3j}^{(k)} \mathbf{Q}_{3j}^{(k)T}\| &= O_{a.s.}\left(\frac{\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2}{d_k} \sqrt{\frac{r_e}{T}}\right), \quad \frac{\|\mathbf{S}_{13}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} = O_{a.s.}\left(\frac{d_k^{\delta_2/2}}{d_k^{1/2}} \left(\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2\right)^{1/2}\right), \\ \frac{\|\mathbf{S}_{12}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} &= O_{a.s.}\left(\frac{T^\delta d_k^{1/2+\delta_2/2} r_k^{1/2}}{d_k^{1/2}}\right) \text{ or } O_P\left(\frac{d_k^{1/2+\delta_2/2} r_k^{1/2}}{T^{1/2} d_k^{1/2}}\right), \\ \|\mathbf{S}_{23}^{(k)}\| &= O_{a.s.}\left(\frac{T^\delta d_k^{1/2} \|\Psi^{(k)}\|_F}{d_k}\right) \text{ or } O_P\left(\frac{d_k^{1/2} \|\Psi^{(k)}\|_F}{T^{1/2} d_k}\right). \end{aligned}$$

If in addition (E3) holds, and (R1), (R2) are replaced by (R1)' and (R2)' respectively, then

$$\begin{aligned} \|\mathbf{S}_{22}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{2j}^{(k)} \mathbf{Q}_{2j}^{(k)\top}\| &= O_{a.s.} \left(\sqrt{\frac{d_k}{Td_k}} + \frac{1}{T} \right), \\ \|\mathbf{S}_{23}^{(k)}\| &= O_{a.s.} \left(\min \left\{ \frac{T^\delta d_k^{1/2} \|\boldsymbol{\Psi}^{(k)}\|_F}{d_k}, \left(\frac{d_k}{Td_k} \right)^{1/4} \right\} \right) \text{ or } O_P \left(\min \left\{ \frac{d_k^{1/2} \|\boldsymbol{\Psi}^{(k)}\|_F}{T^{1/2} d_k}, \left(\frac{d_k}{Td_k} \right)^{1/4} \right\} \right), \\ \frac{\|\mathbf{S}_{12}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} &= O_{a.s.} \left(d_k^{\delta_2/2} \min \left\{ \frac{T^\delta d_k^{1/2} r_k^{1/2}}{d_k^{1/2}}, \left(\frac{d_k}{Td_k} \right)^{1/4} \right\} \right) \text{ or } O_P \left(d_k^{\delta_2/2} \min \left\{ \frac{d_k^{1/2} r_k^{1/2}}{T^{1/2} d_k^{1/2}}, \left(\frac{d_k}{Td_k} \right)^{1/4} \right\} \right), \end{aligned}$$

while other terms have the same rates as stated before.

Proof of Lemma 3. Using (8.22), defining $I = \|T^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} - \mathbf{I}_r\|$,

$$\begin{aligned} \frac{\|\mathbf{S}_{11}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top}\|}{\lambda_j(\mathbf{R}^{(k)})} &= \frac{\|\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} (T^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} - \mathbf{I}_r) \mathbf{Q}_{1j}^{(k)\top}\|}{\lambda_j(\mathbf{R}^{(k)})} \\ &\leq \frac{\sum_{j=1}^{d_k} \|\mathbf{Q}_{1j}^{(k)}\|^2 \cdot \|T^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top} - \mathbf{I}_r\|}{\lambda_j(\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} \\ &= \frac{\|\mathbf{A}_{-k}\|_F^2 \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|^2 \cdot I}{\lambda_j^2(\|\mathbf{A}_{-k}\|_F \|\text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|)} \\ &\leq \|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\| \cdot \|\{\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\}^{-1}\| \cdot I = O(d_k^{\delta_2} I), \quad (8.24) \end{aligned}$$

where the last equality used Assumption (M1). To apply Lemma 1 with $n = (m+1)T$, $X_n = \mathbf{Z}_f^{(k)}$, $B_n = \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top$, we check the following. By Assumption (R2), for m a constant,

- i $\|B_n\| \leq \|\mathbf{A}_{f,T}\|^2$ which is uniformly bounded above;
- ii $T^{-1} \text{tr}(B_n) = T^{-1} \text{tr}(\mathbf{A}_{f,T}^\top \mathbf{A}_{f,T}) - T^{-2} \mathbf{1}_T \mathbf{A}_{f,T}^\top \mathbf{A}_{f,T} \mathbf{1}_T = 1 - o(T^{-3} d^{-2}) - O(T^{-1/2}) \rightarrow 1$;
- iii $T^{-1} \text{tr}(B_n^2) = T^{-1} \text{tr}(\mathbf{A}_{f,T}^\top \mathbf{A}_{f,T})^2 - 2T^{-2} \mathbf{1}^\top (\mathbf{A}_{f,T}^\top \mathbf{A}_{f,T})^2 \mathbf{1}_T + T^{-3} (\mathbf{1}_T \mathbf{A}_{f,T}^\top \mathbf{A}_{f,T} \mathbf{1}_T)^2 \rightarrow a_f - 2b_f + c_f^2$.

Hence (ii) and (iii) imply $n^{-1} \text{tr}(B_n)$ and $n^{-1} \text{tr}(B_n^2)$ approach certain limits as $T \rightarrow \infty$. Lemma 1 then implies that for each $j \in [r_k]$,

$$\sqrt{(m+1)T/r} \{ \lambda_j([(m+1)T]^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) - [(m+1)T]^{-1} \text{tr}(\mathbf{A}_{f,T}^\top \mathbf{A}_{f,T} \mathbf{M}_T) \} = O_{a.s.}(1).$$

This implies that, since m is a constant, we have for $j \in [r]$,

$$\lambda_j(T^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) - 1 + o(T^{-3} d^{-2}) + O(T^{-1/2}) = O_{a.s.}(r^{1/2} T^{-1/2}), \quad (8.25)$$

which implies that in particular

$$I = \|T^{-1}\mathbf{Z}_f^{(k)}\mathbf{A}_{f,T}\mathbf{M}_T\mathbf{A}_{f,T}^\top\mathbf{Z}_f^{(k)\top} - \mathbf{I}_r\| = O_{a.s.}(r^{1/2}T^{-1/2}). \quad (8.26)$$

Hence combining with (8.24), we have

$$\frac{\|\mathbf{S}_{11}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)}\mathbf{Q}_{1j}^{(k)\top}\|}{\lambda_j(\mathbf{R}^{(k)})} = O_{a.s.}\left(d_k^{\delta_2}\sqrt{\frac{r}{T}}\right).$$

For $\mathbf{S}_{22}^{(k)}$, we first analyse without Assumption (E3). Define $I_j = \|T^{-1}\mathbf{Z}_{\epsilon,j}^{(k)}\mathbf{A}_{\epsilon,T}\mathbf{M}_T\mathbf{A}_{\epsilon,T}^\top\mathbf{Z}_{\epsilon,j}^{(k)\top} - \mathbf{I}_{d_k}\|$.

Then using (8.23),

$$\|\mathbf{S}_{22}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{2j}^{(k)}\mathbf{Q}_{2j}^{(k)\top}\| \leq \sum_{j=1}^{d_k} \|\mathbf{Q}_{2j}^{(k)}\|^2 \cdot I_j = O(\max_{j \in [d_k]} I_j).$$

Using (4.17) and (4.18) of Wang and Paul (2014), we have for each $k \in [K]$,

$$P(\|T^{-1}\mathbf{Z}_{\epsilon,j}^{(k)}\mathbf{A}_{\epsilon,T}\mathbf{M}_T\mathbf{A}_{\epsilon,T}^\top\mathbf{Z}_{\epsilon,j}^{(k)\top} - T^{-1}\text{tr}(\mathbf{A}_{\epsilon,T}^\top\mathbf{A}_{\epsilon,T}\mathbf{M}_t) \cdot \mathbf{I}_{d_k}\| > C_\eta d_k^{1/2}T^{-1/2}) \leq \exp(-\eta d_k),$$

where $\eta > 0$ and $C_\eta > 0$ is a constant independent of T . Since $\mathbf{1}_T^\top\mathbf{A}_{\epsilon,T}^\top\mathbf{A}_{\epsilon,T}\mathbf{1}_T/T^2 = O(T^{-1/2})$ by Assumption (R2), we have for a constant $C'_\eta > 0$, and for each $j \in [d_k]$,

$$P(I_j > C'_\eta d_k^{1/2}T^{-1/2}) \leq \exp(-\eta d_k).$$

The union-sum inequality then implies that

$$P(\max_{j \in [d_k]} I_j > C'_\eta d_k^{1/2}T^{-1/2}) \leq d_k \exp(-\eta d_k) = o(T^{K-1} \exp(-\eta T^c)),$$

where the last equality used $d_k/T = o(1)$ and $T^c = O(d_k)$ from Assumption (R1). Since $K = o(T^c/\log T)$ by Assumption (R1), Borel-Cantelli's lemma implies that $\max_{j \in [d_k]} I_j = O_{a.s.}(d_k^{1/2}T^{-1/2})$.

For $\mathbf{S}_{33}^{(k)}$, define $J = \|T^{-1}\mathbf{Z}_e^{(k)}\mathbf{A}_{e,T}\mathbf{M}_T\mathbf{A}_{e,T}^\top\mathbf{Z}_e^{(k)\top} - \mathbf{I}_{r_e}\|$. Then by (8.22),

$$\|\mathbf{S}_{33}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{3j}^{(k)}\mathbf{Q}_{3j}^{(k)\top}\| \leq \|\text{diag}(d_k^{-1}\mathbf{\Sigma}^{(k)})^{-1}\| \cdot d_k^{-1} \sum_{j=1}^{d_k} \|\mathbf{\Psi}_j^{(k)}\|^2 \cdot J = O\left(d_k^{-1} \sum_{j=1}^{d_k} \|\mathbf{\Psi}_j^{(k)}\|^2 \cdot J\right).$$

Similar to the application of Lemma 1 in the treatment of $\mathbf{S}_{11}^{(k)}$ at the beginning of the proof, we can set $n = (m+1)T$, $X_n = \mathbf{Z}_e^{(k)}$ and $B_n = A_{e,T}\mathbf{M}_T\mathbf{A}_{e,T}^\top$, and hence (details of applicability checks similar to

that for $\mathbf{S}_{11}^{(k)}$ and is omitted)

$$\sqrt{(m+1)T/r_e} \{ \lambda_j ([(m+1)T]^{-1} \mathbf{Z}_e^{(k)} \mathbf{A}_{e,T} \mathbf{M}_T \mathbf{A}_{e,T}^\top \mathbf{Z}_e^{(k)\top}) - [(m+1)T]^{-1} \text{tr}(\mathbf{A}_{e,T}^\top \mathbf{A}_{e,T} \mathbf{M}_T) \} = O_{a.s.}(1).$$

Using Assumption (R2), the above implies that $J = O_{a.s.}(r_e^{1/2} T^{-1/2})$, and hence $\|\mathbf{S}_{33}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{3j}^{(k)} \mathbf{Q}_{ej}^{(k)\top}\|$ has rate as stated in the lemma.

For $\mathbf{S}_{21}^{(k)}$, define $I_{21} = \|T^{-1} d_{-k}^{-1/2} \sum_{j=1}^{d_k} (\boldsymbol{\Sigma}_{\epsilon,j}^{(k)})^{1/2} \mathbf{Z}_{\epsilon,j}^{(k)} \mathbf{A}_{\epsilon,T} \mathbf{M}_T \mathbf{A}_{\epsilon,T}^\top \mathbf{Z}_f^{(k)\top} (\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j)\|_F$. Then

$$EI_{21}^2 = T^{-2} d_{-k}^{-1} \sum_{j=1}^{d_k} \text{tr}(\boldsymbol{\Sigma}_{\epsilon,j}^{(k)}) \text{tr}(\mathbf{A}_{\epsilon,T}^\top \mathbf{A}_{\epsilon,j} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{A}_{f,T} \mathbf{M}_T) \cdot r_k \|(\mathbf{A}_{-k}^\top)_j\|^2 = O(T^{-1} d_{-k}^{-1} d_k r_k \|\mathbf{A}_{-k}\|_F^2)$$

by Assumption (E1) and (R2). Hence $I_{21} = O_{a.s.}(T^\delta d_k^{1/2} r_k^{1/2} d_{-k}^{-1/2} \|\mathbf{A}_{-k}\|_F)$ for some small $\delta > 0$, or we have $I_{21} = O_P(d_k^{1/2} r_k^{1/2} T^{-1/2} d_{-k}^{-1/2} \|\mathbf{A}_{-k}\|_F)$. Then

$$\begin{aligned} \frac{\|\mathbf{S}_{21}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} &\leq \frac{\|\text{diag}(d_{-k}^{-1} \boldsymbol{\Sigma}^{(k)})^{-1/2}\| \cdot I_{21} \cdot \|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\|^{1/2}}{\lambda_j^{1/2}(\mathbf{R}^{(k)}) \lambda_j^{1/2} (\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} \\ &= O(I_{21} \cdot \|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\|^{1/2} \cdot \|\mathbf{A}_{-k}\|_F^{-1} \|\{\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\}^{-1}\|^{1/2}) \\ &= O_{a.s.} \left(\frac{T^\delta r_k^{1/2} d_k^{1/2+\delta_2/2}}{d_{-k}^{1/2}} \right) \text{ or } O_P \left(\frac{r_k^{1/2} d_k^{1/2+\delta_2/2}}{T^{1/2} d_{-k}^{1/2}} \right). \end{aligned}$$

For $\mathbf{S}_{31}^{(k)}$, define $I_{31} = \|T^{-1} \mathbf{Z}_e^{(k)} \mathbf{A}_{e,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}\|$. Then we have

$$\begin{aligned} I_{31} &\leq \|T^{-1} \mathbf{Z}_e^{(k)} \mathbf{Z}_e^{(k)\top}\|^{1/2} \cdot \|T^{-1} \mathbf{Z}_f^{(k)} \mathbf{Z}_f^{(k)\top}\|^{1/2} \|\mathbf{A}_{e,T}\| \|\mathbf{A}_{f,T}\| \\ &= O_{a.s.} \{ (m+1) \cdot (1 + r_e^{1/2} [(m+1)T]^{-1/2}) \cdot (1 + r^{1/2} [(m+1)T]^{-1/2}) \} = O_{a.s.}(1), \end{aligned}$$

where we used Assumption (R1), and Theorem 5.8 of Bai and Silverstein (2010) on the almost sure limit of the largest eigenvalue of a sample covariance matrix. Hence

$$\begin{aligned} \frac{\|\mathbf{S}_{31}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} &\leq \frac{\|\text{diag}(d_{-k}^{-1} \boldsymbol{\Sigma}^{(k)})^{-1/2}\| \cdot d_{-k}^{-1/2} \sum_{j=1}^{d_k} \|\boldsymbol{\Psi}_j^{(k)}\| \|\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j\| \cdot I_{31} \cdot \|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\|^{1/2}}{\lambda_j^{1/2}(\mathbf{R}^{(k)}) \lambda_j^{1/2} (\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} \\ &\leq O \left(d_{-k}^{-1/2} \left(\sum_{j=1}^{d_k} \|\boldsymbol{\Psi}_j^{(k)}\|^2 \right)^{1/2} \right) \|\mathbf{A}_{-k}\|_F \cdot I_{31} \cdot (\|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\| \|\{\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\}^{-1}\|)^{1/2} \|\mathbf{A}_{-k}\|_F^{-1} \\ &= O_{a.s.} \left(\frac{d_k^{\delta_2/2}}{d_{-k}^{1/2}} \left(\sum_{j=1}^{d_k} \|\boldsymbol{\Psi}_j^{(k)}\|^2 \right)^{1/2} \right). \end{aligned}$$

For $\mathbf{S}_{32}^{(k)}$, define $I_{32} = \|T^{-1} d_{-k}^{-1} \sum_{j=1}^{d_k} (\boldsymbol{\Sigma}_{\epsilon,j}^{(k)})^{1/2} \mathbf{Z}_{\epsilon,j}^{(f)} \mathbf{A}_{\epsilon,T} \mathbf{M}_T \mathbf{A}_{e,T}^\top \mathbf{Z}_e^{(k)\top} \boldsymbol{\Psi}_j^{(k)\top}\|_F$. Then by Assumption

(R1) and (R2),

$$EI_{32}^2 = T^{-2} d_{-k}^{-2} \sum_{j=1}^{d_{-k}} \text{tr}(\Sigma_{\epsilon,j}^{(k)}) \text{tr}(\Psi_j^{(k)} \Psi_j^{(k)\top}) \text{tr}(\mathbf{A}_{\epsilon,T}^\top \mathbf{A}_{\epsilon,T} \mathbf{M}_T \mathbf{A}_{\epsilon,T}^\top \mathbf{A}_{\epsilon,T} \mathbf{M}_T) = O(T^{-1} d_{-k}^{-2} d_k \|\Psi^{(k)}\|_F^2),$$

so that $I_{32} = O_{a.s.}(T^\delta d_{-k}^{-1} d_k^{1/2} \|\Psi^{(k)}\|_F)$ for some small constant $\delta > 0$, or $O_P(T^{-1/2} d_{-k}^{-1} d_k^{1/2} \|\Psi^{(k)}\|_F)$, which is also the rate for $\|\mathbf{S}_{32}^{(k)}\|$.

Now consider Assumption (E3) in place and (R1) and (R2) replaced by (R1)' and (R2)' respectively. We can write

$$\mathbf{S}_{21}^{(k)} = \text{diag}(d_{-k}^{-1} \Sigma^{(k)})^{-1/2} \cdot T^{-1} d_{-k}^{-1/2} \mathbf{D}_\sigma \mathbf{Z}^{(k)} \mathbf{G} \cdot \mathbf{A}_k^\top \text{diag}(\Sigma^{(k)})^{-1/2}, \quad \text{where } \mathbf{D}_\sigma := \text{diag}(\sigma_1^{(k)}, \dots, \sigma_{d_k}^{(k)}),$$

$$\mathbf{Z}^{(k)} := (\mathbf{Z}_{\epsilon,1}^{(k)}, \dots, \mathbf{Z}_{\epsilon,d_{-k}}^{(k)}), \quad \mathbf{G} := (\mathbf{I}_{d_{-k}} \otimes \mathbf{A}_{\epsilon,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) \begin{pmatrix} \mathbf{I}_{r_k} \otimes w_1^{(k)} (\mathbf{A}_{-k}^\top)_1 \\ \vdots \\ \mathbf{I}_{r_k} \otimes w_{d_{-k}}^{(k)} (\mathbf{A}_{-k}^\top)_{d_{-k}} \end{pmatrix}.$$

Define $\mathbf{J}_{21} := T^{-1} d_{-k}^{-1/2} \|\mathbf{A}_{-k}\|_F^{-1} \mathbf{Z}^{(k)} \mathbf{G}$. Firstly, we have

$$\begin{aligned} T^{-1} \|\mathbf{A}_{-k}\|_F^{-2} \|\mathbf{G}^\top \mathbf{G}\| &\leq \|\mathbf{A}_{-k}\|_F^{-2} \sum_{j=1}^{d_{-k}} w_j^{(k)2} \|\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j^\top\|^2 \cdot \|T^{-1/2} \mathbf{Z}_f^{(k)\top} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{\epsilon,T}^\top\|^2 \\ &\leq C^2 \|\mathbf{A}_{-k}\|_F^{-2} \|\mathbf{A}_{-k}\|_F^2 \cdot \|T^{-1} \mathbf{Z}_f^{(k)} \mathbf{Z}_f^{(k)\top}\| = O_{a.s.}(1), \end{aligned}$$

where the second inequality used Assumption (E3) and (R2), and the last equality used Theorem 5.8 of Bai and Silverstein (2010). Hence defining $\mathbf{B} := T^{-1} \|\mathbf{A}_{-k}\|_F^{-2} \mathbf{G} \mathbf{G}^\top$, we can, on an almost sure set, apply (8.17), so that $\mathbf{J}_{21} \mathbf{J}_{21}^\top = T^{-1} d_{-k}^{-1} \mathbf{Z}^{(k)} \mathbf{B} \mathbf{Z}^{(k)\top}$, and for $j \in [d_k]$,

$$\sqrt{(m+1)Td_{-k}/d_k} (\lambda_j((m+1)^{-1} \mathbf{J}_{21} \mathbf{J}_{21}^\top) - [(m+1)Td_{-k}]^{-1} \text{tr}(\mathbf{B})) = O_{a.s.}(1).$$

This implies that for $j \in [d_k]$,

$$\lambda_j(\mathbf{J}_{21} \mathbf{J}_{21}^\top) = O_{a.s.} \left(\sqrt{\frac{d_k}{Td_{-k}}} + \frac{\text{rank}(\mathbf{B}) \|\mathbf{B}\|}{Td_{-k}} \right) = O_{a.s.} \left(\sqrt{\frac{d_k}{Td_{-k}}} + \frac{r}{Td_{-k}} \right) = O_{a.s.} \left(\sqrt{\frac{d_k}{Td_{-k}}} \right),$$

where we used $r = o(T)$ and $r_k = o(d_k)$ for each $k \in [K]$ in Assumption (M1). Hence for $j \in [r_k]$,

$$\frac{\|\mathbf{S}_{21}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} = \frac{\|\text{diag}(d_{-k}^{-1} \Sigma^{(k)})^{1/2}\| \|\mathbf{D}_\sigma\| \|\mathbf{J}_{21}\| \|\mathbf{A}_{-k}\|_F \|\mathbf{A}_k^\top \text{diag}(\Sigma^{(k)})^{-1} \mathbf{A}_k\|^{1/2}}{\lambda_j^{1/2}(\mathbf{R}^{(k)}) \|\mathbf{A}_{-k}\|_F \lambda_j^{1/2}(\mathbf{A}_k^\top \text{diag}(\Sigma^{(k)})^{-1} \mathbf{A}_k)} = O_{a.s.} \left(d_k^{\delta_2/2} \left(\frac{d_k}{Td_{-k}} \right)^{1/4} \right).$$

Combining with the result from before for $\mathbf{S}_{21}^{(k)}$ without Assumption (E3), which is in fact still valid under

Assumption (E3), (R1) and (R2), we thus have

$$\frac{\|\mathbf{S}_{21}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} = O_{a.s.} \left(d_k^{\delta_2/2} \min \left\{ \frac{T^\delta d_k^{1/2} r_k^{1/2}}{d_k^{1/2}}, \left(\frac{d_k}{T d_k} \right)^{1/4} \right\} \right) \text{ or } O_P \left(d_k^{\delta_2/2} \min \left\{ \frac{d_k^{1/2} r_k^{1/2}}{T^{1/2} d_k^{1/2}}, \left(\frac{d_k}{T d_k} \right)^{1/4} \right\} \right).$$

Define $\mathbf{G}_w := \text{diag}(w_1^{(k)2}, \dots, w_{d_k}^{(k)2}) \otimes \mathbf{A}_{\epsilon, T} \mathbf{M}_T \mathbf{A}_{\epsilon, T}^\top$, $J_{22} := \|T^{-1} d_k^{-1} \mathbf{Z}^{(k)} \mathbf{G}_w \mathbf{Z}^{(k)\top} - d_k^{-1} \sum_{j=1}^{d_k} w_j^{(k)2} \mathbf{I}_{d_k}\|$, then since $\|\mathbf{G}_w\| \leq \|\mathbf{A}_{\epsilon, T}\|^2 \max_{j \in [d_k]} w_j^{(k)2} < \infty$ by Assumption (E3) and (R2), we apply (8.17) to conclude that for $j \in [d_k]$,

$$\sqrt{(m+1)T d_k / d_k} \{ \lambda_j((m+1)T d_k)^{-1} \mathbf{Z}^{(k)} \mathbf{G}_w \mathbf{Z}^{(k)\top} - [(m+1)T d_k]^{-1} \text{tr}(\mathbf{G}_w) \} = O_{a.s.}(1).$$

But by Assumption (R2)',

$$T^{-1} d_k^{-1} \text{tr}(\mathbf{G}_w) = d_k^{-1} \sum_{j=1}^{d_k} w_j^{(k)2} \cdot T^{-1} \text{tr}(\mathbf{A}_{\epsilon, T}^\top \mathbf{A}_{\epsilon, T} \mathbf{M}_T) = d_k^{-1} \sum_{j=1}^{d_k} w_j^{(k)2} (1 - o(T^{-3} d_k^{-2}) + O(T^{-1})),$$

and hence the above implies that

$$\|\mathbf{S}_{22}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{2j}^{(k)} \mathbf{Q}_{2j}^{(k)\top}\| \leq \|\text{diag}(d_k^{-1} \boldsymbol{\Sigma}^{(k)})^{-1}\| \|\mathbf{D}_\sigma\|^2 \cdot J_{22} = O_{a.s.}(J_{22}) = O_{a.s.} \left(\sqrt{\frac{d_k}{T d_k}} + \frac{1}{T} \right).$$

Define $\mathbf{J}_{23} = T^{-1} d_k^{-1} \mathbf{Z}^{(k)} \mathbf{G}_\psi$, where $\mathbf{G}_\psi := (\mathbf{I}_{d_k} \otimes \mathbf{A}_{\epsilon, T} \mathbf{M}_T \mathbf{A}_{\epsilon, T}^\top \mathbf{Z}_e^{(k)\top}) \text{diag}(w_1^{(k)}, \dots, w_{d_k}^{(k)}) \boldsymbol{\Psi}^{(k)\top}$. Hence $\|T^{-1/2} d_k^{-1/2} \mathbf{G}_\psi\| = \|T^{-1/2} \mathbf{Z}_e^{(k)}\| \|\mathbf{A}_{\epsilon, T} \mathbf{M}_T \mathbf{A}_{\epsilon, T}^\top\| \cdot \max_{j \in [d_k]} w_j^{(k)} \cdot \|d_k^{-1/2} \boldsymbol{\Psi}^{(k)}\| = O_{a.s.}(1)$ by Assumption (M1), (E3), (R2), and Theorem 5.8 of Bai and Silverstein (2010). Applying (8.17), for $j \in [d_k]$,

$$\sqrt{(m+1)T d_k / d_k} \{ \lambda_j((m+1)^{-1} \mathbf{J}_{23} \mathbf{J}_{23}^\top) - [(m+1)T d_k]^{-1} \text{tr}(T^{-1} d_k^{-1} \mathbf{G}_\psi \mathbf{G}_\psi^\top) \} = O_{a.s.}(1).$$

Since $\text{rank}(\mathbf{G}_\psi) = d_k$, the above implies $\lambda_j(\mathbf{J}_{23} \mathbf{J}_{23}^\top) = O_{a.s.}\{(d_k / (T d_k))^{1/2}\}$. Hence

$$\|\mathbf{S}_{23}^{(k)}\| = \|\text{diag}(d_k^{-1} \boldsymbol{\Sigma}^{(k)})^{-1}\| \|\mathbf{D}_\sigma\| \|\mathbf{J}_{23}\| = O_{a.s.} \left(\frac{d_k}{T d_k} \right)^{1/4}.$$

Combining with the result for $\mathbf{S}_{23}^{(k)}$ before which is in fact applicable with Assumption (E3), (R1) and (R2), we have the rate as stated in the statement of the lemma, and the proof of the lemma completes. $\square$

Lemma 4. *Let Assumption (E1), (E2), (F1), (M1), (R1) and (R2) hold. Then for a small constant $\delta > 0$,*

$$\begin{aligned}
& \max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| \\
&= O_{a.s.} \left(d_k^{\delta_2} \sqrt{\frac{r}{T}} + \sqrt{\frac{d_k}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} + \frac{T^\delta d_k^{1/2}}{d_{-k}^{1/2}} \left(d_k^{\delta_2/2} r_k^{1/2} + \frac{\|\Psi^{(k)}\|_F}{d_{-k}^{1/2}} \right) \right) \quad \text{or} \\
&= O_P \left(d_k^{\delta_2} \sqrt{\frac{r}{T}} + \sqrt{\frac{d_k}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} + \frac{d_k^{1/2}}{T^{1/2} d_{-k}^{1/2}} \left(d_k^{\delta_2/2} r_k^{1/2} + \frac{\|\Psi^{(k)}\|_F}{d_{-k}^{1/2}} \right) \right). \quad \text{And,} \\
& \max_{j \in [d_k] \setminus [r_k]} |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})| = O_{a.s.} \left(\sqrt{\frac{d_k}{T}} + \frac{T^\delta d_k^{1/2} \|\Psi^{(k)}\|_F}{d_{-k}} \right) \quad \text{or} \quad O_P \left(\sqrt{\frac{d_k}{T}} + \frac{d_k^{1/2} \|\Psi^{(k)}\|_F}{T^{1/2} d_{-k}} \right),
\end{aligned}$$

with the latter convergence in probability holds with (R2) replaced by (R2P).

If in addition (E3) holds, and (R1), (R2) are replaced by (R1)' and (R2)' respectively, then

$$\begin{aligned}
& \max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| = O_{a.s.} \left(d_k^{\delta_2} \sqrt{\frac{r}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} + \right. \\
& \quad \left. d_k^{\delta_2/2} \min \left\{ \frac{T^\delta d_k^{1/2} r_k^{1/2}}{d_{-k}^{1/2}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} + \min \left\{ \frac{T^\delta d_k^{1/2} \|\Psi^{(k)}\|_F}{d_{-k}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} \right) \quad \text{or} \\
& \quad = O_P \left(d_k^{\delta_2} \sqrt{\frac{r}{T}} + \frac{d_k^{\delta_2/2} (\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2)^{1/2}}{d_{-k}^{1/2}} + \right. \\
& \quad \left. d_k^{\delta_2/2} \min \left\{ \frac{d_k^{1/2} r_k^{1/2}}{T^{1/2} d_{-k}^{1/2}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} + \min \left\{ \frac{d_k^{1/2} \|\Psi^{(k)}\|_F}{T^{1/2} d_{-k}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} \right). \quad \text{And,} \\
& \max_{j \in [d_k] \setminus [r_k]} |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})| \\
&= O_{a.s.} \left(\sqrt{\frac{d_k}{T d_{-k}}} + \frac{1}{T} + \frac{\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2}{d_{-k}} \sqrt{\frac{r_e}{T}} + \min \left\{ \frac{T^\delta d_k^{1/2} \|\Psi^{(k)}\|_F}{d_{-k}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} \right) \quad \text{or} \\
&= O_P \left(\sqrt{\frac{d_k}{T d_{-k}}} + \frac{1}{T} + \frac{\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2}{d_{-k}} \sqrt{\frac{r_e}{T}} + \min \left\{ \frac{d_k^{1/2} \|\Psi^{(k)}\|_F}{T^{1/2} d_{-k}}, \left(\frac{d_k}{T d_{-k}} \right)^{1/4} \right\} \right),
\end{aligned}$$

with the latter convergence in probability holds with (R2)' replaced by (R2P)'.

The rate for $\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|$ is the same as that for $\max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|$ except for setting $\delta_2 = 0$ in both scenarios.

Proof of Lemma 4. We prove the first part of the lemma first without Assumption (E3). Firstly, for

$j \in [r_k]$, using $\lambda_j(\mathbf{R}^{(k)}) > 1$ from Theorem 1,

$$\begin{aligned} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| &\leq \frac{\|\mathbf{S}^{(k)} - \mathbf{R}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} \leq \frac{\sum_{i=1}^3 \|\mathbf{S}_{ii}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}\| + \sum_{i \neq j} \|\mathbf{S}_{ij}^{(k)}\| + \sum_{i \neq j} \|\check{\mathbf{S}}_{ij}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} \\ &\leq \frac{\|\mathbf{S}_{11}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top}\| + 2\|\mathbf{S}_{12}^{(k)}\| + 2\|\mathbf{S}_{13}^{(k)}\|}{\lambda_j(\mathbf{R}^{(k)})} + \sum_{i=2}^3 \|\mathbf{S}_{ii}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}\| + 2\|\mathbf{S}_{23}^{(k)}\| + \sum_{i \neq j} \|\check{\mathbf{S}}_{ij}^{(k)}\|. \end{aligned} \quad (8.27)$$

Using the results from Lemma 2 and Lemma 3, noticing that the rate for $\|\mathbf{S}_{33}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{3j}^{(k)} \mathbf{Q}_{3j}^{(k)\top}\|$ is dominated by that of $\|\mathbf{S}_{13}^{(k)}\|/\lambda_j(\mathbf{R}^{(k)})$, we have the result for $\max_{j \in [r_k]} |\lambda_j(\mathbf{S}^{(k)})/\lambda_j(\mathbf{R}^{(k)}) - 1|$.

Now consider for $j \in [d_k]$,

$$\lambda_j(\mathbf{S}^{(k)} \text{diag}(\mathbf{S}^{(k)})^{-1}) \lambda_{d_k}(\text{diag}(\mathbf{S}^{(k)})) \leq \lambda_j(\mathbf{S}^{(k)}) \leq \lambda_j(\mathbf{S}^{(k)} \text{diag}(\mathbf{S}^{(k)})^{-1}) \lambda_1(\text{diag}(\mathbf{S}^{(k)})),$$

so that, using $\widehat{\mathbf{R}}^{(k)} = \text{diag}(\mathbf{S}^{(k)})^{-1/2} \mathbf{S}^{(k)} \text{diag}(\mathbf{S}^{(k)})^{-1/2}$ and defining $\mathbf{u}_j$ to be a unit vector with 1 at the j th position and 0 elsewhere,

$$\begin{aligned} \max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}^{(k)})} - 1 \right| &= \max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{S}^{(k)} \text{diag}(\mathbf{S}^{(k)})^{-1})} - 1 \right| \leq \max_{j \in [d_k]} |\lambda_j(\text{diag}(\mathbf{S}^{(k)})) - 1| \\ &= \max_{j \in [d_k]} |\lambda_j(\text{diag}(\mathbf{S}^{(k)} - \mathbf{R}^{(k)}))| = \max_{j \in [d_k]} |\mathbf{u}_j^\top (\mathbf{S}^{(k)} - \mathbf{R}^{(k)}) \mathbf{u}_j| \\ &\leq \max_{j \in [d_k]} \left| \mathbf{u}_j^\top \left(\mathbf{S}_{11}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top} \right) \mathbf{u}_j \right| + 2 \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{12}^{(k)} \mathbf{u}_j| + 2 \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{13}^{(k)} \mathbf{u}_j| \\ &\quad + \sum_{i=2}^3 \|\mathbf{S}_{ii}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}\| + 2\|\mathbf{S}_{23}^{(k)}\| + \sum_{i \neq j} \|\check{\mathbf{S}}_{ij}^{(k)}\|. \end{aligned} \quad (8.28)$$

Consider $I = \|\mathbf{T}^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_{f,T}^{(k)} - \mathbf{I}_r\| = O_{a.s.}(r^{1/2} T^{-1/2})$ by (8.26). Then

$$\begin{aligned} \max_{j \in [d_k]} \left| \mathbf{u}_j^\top \left(\mathbf{S}_{11}^{(k)} - \sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top} \right) \mathbf{u}_j \right| &\leq \|\mathbf{A}_{-k}\|_F^{-2} \sum_{j=1}^{d_k} \|\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j^\top\|^2 \cdot I \\ &\quad \cdot \max_{j \in [d_k]} \|\mathbf{u}_j^\top \text{diag}(\|\mathbf{A}_{-k}\|_F^{-2} \mathbf{\Sigma}^{(k)})^{-1/2} \mathbf{A}_{-k}\|^2 \leq I = O_{a.s.}(r^{1/2} T^{-1/2}), \end{aligned}$$

where the last inequality used $\|\mathbf{A}_{-k}\|_F^2 = \sum_{j=1}^{d_k} \|\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j^\top\|^2$, and $\|\mathbf{u}_j^\top \text{diag}(\|\mathbf{A}_{-k}\|_F^{-2} \mathbf{\Sigma}^{(k)})^{-1/2} \mathbf{A}_{-k}\|^2 \leq 1$ by the definition of $\mathbf{\Sigma}^{(k)}$.

Define $I_{21} = \|T^{-1} d_{-k}^{-1/2} \sum_{j=1}^{d_k} (\mathbf{\Sigma}_{\epsilon,j}^{(k)})^{1/2} \mathbf{Z}_{\epsilon,j}^{(k)} \mathbf{A}_{\epsilon,T} \mathbf{M}_T \mathbf{A}_{\epsilon,T}^\top \mathbf{Z}_f^{(k)\top} (\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j)\|_F$, which has rate in the proof of Lemma 3 to be $O_{a.s.}(T^\delta (d_k r_k / d_{-k})^{1/2} \|\mathbf{A}_{-k}\|_F)$ for a small $\delta > 0$, or $O_P((d_k r_k / (T d_{-k}))^{1/2} \|\mathbf{A}_{-k}\|_F)$.

Then

$$\begin{aligned} \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{21}^{(k)} \mathbf{u}_j| &\leq \|\text{diag}(d_k^{-1} \boldsymbol{\Sigma}^{(k)})^{-1/2}\| \cdot I_{21} \cdot \|\mathbf{A}_{-k}\|_F^{-1} \cdot \max_{j \in [d_k]} \|\mathbf{u}_j^\top \text{diag}(\|\mathbf{A}_{-k}\|_F^{-2} \boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\| \\ &= O_{a.s.}(T^\delta (d_k r_k / d_{-k})^{1/2}) \quad \text{or} \quad O_P((d_k r_k / (T d_{-k}))^{1/2}). \end{aligned}$$

Also, define $I_{31} = \|T^{-1} \mathbf{Z}_e^{(k)} \mathbf{A}_{e,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}\| = O_{a.s.}(1)$ as in the proof of Lemma 3. Then

$$\begin{aligned} \max_{j \in [d_k]} |\mathbf{u}_j^\top \mathbf{S}_{31}^{(k)} \mathbf{u}_j| &\leq \|\text{diag}(d_k^{-1} \boldsymbol{\Sigma}^{(k)})^{-1/2}\| \cdot d_{-k}^{-1/2} \sum_{j=1}^{d_{-k}} \|\Psi_j^{(k)}\| \|\mathbf{I}_{r_k} \otimes (\mathbf{A}_{-k}^\top)_j^\top\| \cdot I_{31} \\ &\quad \cdot \|\mathbf{A}_{-k}\|_F^{-1} \cdot \max_{j \in [d_k]} \|\mathbf{u}_j^\top \text{diag}(\|\mathbf{A}_{-k}\|_F^{-2} \boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\| = O_{a.s.} \left(\frac{1}{d_{-k}^{1/2}} \left(\sum_{j=1}^{d_{-k}} \|\Psi_j^{(k)}\|^2 \right)^{1/2} \right). \end{aligned}$$

Hence with the three rates above, from (8.28), and inspecting the rate for $\max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|$, we can see that the rate for $\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|$ is the same as that for $\max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|$ except for setting $\delta_2 = 0$.

Now consider $j \in [d_k] \setminus [r_k]$. Then

$$\begin{aligned} |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})| &\leq \left| \lambda_j \left(\mathbf{S}_{11}^{(k)} + \sum_{j=2}^3 (\mathbf{S}_{1j}^{(k)} + \mathbf{S}_{j1}^{(k)}) + \sum_{i=2}^3 \sum_{j=1}^{d_{-k}} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top} \right) - \lambda_j(\mathbf{R}^{(k)}) \right| \\ &\quad + 2 \|\mathbf{S}_{23}^{(k)}\| + \sum_{i=2}^3 \|\mathbf{S}_{ii}^{(k)} - \sum_{j=1}^{d_{-k}} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}\| + \sum_{i \neq j} \|\check{\mathbf{S}}_{ij}^{(k)}\| \\ &\leq 2 \|\mathbf{S}_{23}^{(k)}\| + \sum_{i=2}^3 \|\mathbf{S}_{ii}^{(k)} - \sum_{j=1}^{d_{-k}} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}\| + \sum_{i \neq j} \|\check{\mathbf{S}}_{ij}^{(k)}\|, \end{aligned} \quad (8.29)$$

if we can prove that for $j \in [d_k] \setminus [r_k]$, either almost surely or in probability,

$$\lambda_j \left(\mathbf{S}_{11}^{(k)} + \sum_{j=2}^3 (\mathbf{S}_{1j}^{(k)} + \mathbf{S}_{j1}^{(k)}) + \sum_{i=2}^3 \sum_{j=1}^{d_{-k}} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top} \right) = \lambda_j(\mathbf{R}^{(k)}). \quad (8.30)$$

Then (8.29) together with the results in Lemma 2 and 3 give the rate, as well as the mode of convergence, of $\max_{j \in [d_k]} |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})|$. Hence it remains to show (8.30). To this end, note that $\text{rank}(\mathbf{S}_{11}^{(k)}) = r_k$. Define the null space of $\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2}$ to be

$$\mathcal{N} := \{\mathbf{w} \in \mathbb{R}^{d_k} : \mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{w} = \mathbf{0}\},$$

and $\mathcal{N}^\perp$ be the corresponding row space. Observe that the dimension of $\mathcal{N}$ is $d_k - r_k$ by Assumption (M1)

that each $\mathbf{A}_k$ is of full rank. Hence the column space of $\mathbf{S}_{11}^{(k)}$ has dimension at most r_k . Consider

$$\begin{aligned} \frac{\lambda_{r_k}(\mathbf{S}_{11}^{(k)})}{\lambda_1(\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} &\geq \frac{\min_{\mathbf{w} \in \mathcal{N}^\perp} \|\mathbf{w}^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1/2} \mathbf{A}_k\|^2 \lambda_r(T^{-1} \mathbf{Z}_f^{(k)} \mathbf{A}_{f,T} \mathbf{M}_T \mathbf{A}_{f,T}^\top \mathbf{Z}_f^{(k)\top}) \|\mathbf{A}_k\|_F^2}{\lambda_1(\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} \\ &= \frac{\lambda_r(\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k) \cdot (1 + o_{a.s.}(1))}{\|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\|} \\ &= \frac{1 + o_{a.s.}(1)}{\|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\| \|\{\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\}^{-1}\|} \geq c(1 + o_{a.s.}(1)) d_k^{-\delta_2}, \end{aligned}$$

where the first equality is by (8.25), and the last inequality by Assumption (M1). But with exactly the same proofs as in Lemma 3, we have

$$\begin{aligned} \frac{\|\mathbf{S}_{12}^{(k)}\|}{\lambda_1(\mathbf{R}^{(k)})} &\leq \frac{\|\mathbf{S}_{12}^{(k)}\|}{\lambda_1(\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} = O_{a.s.}\left(\frac{T^\delta d_k^{1/2+\delta_2/2} r_k^{1/2}}{d_k^{1/2}}\right), \text{ or } O_P\left(\frac{d_k^{1/2+\delta_2/2} r_k^{1/2}}{T^{1/2} d_k^{1/2}}\right) = o(d_k^{-\delta_2}), \\ \frac{\|\mathbf{S}_{13}^{(k)}\|}{\lambda_1(\mathbf{R}^{(k)})} &= \frac{\|\mathbf{S}_{13}^{(k)}\|}{\lambda_1(\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} = O_{a.s.}\left(\frac{d_k^{\delta_2/2}}{d_k^{1/2}} \left(\sum_{j=1}^{d_k} \|\Psi_j^{(k)}\|^2\right)^{1/2}\right) = o(d_k^{-\delta_2}), \\ \frac{\|\sum_{i=2}^3 \sum_{j=1}^{d_k} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}\|}{\lambda_1(\sum_{j=1}^{d_k} \mathbf{Q}_{1j}^{(k)} \mathbf{Q}_{1j}^{(k)\top})} &\leq \frac{1}{\|\mathbf{A}_k\|_F^2 \|\mathbf{A}_k^\top \text{diag}(\boldsymbol{\Sigma}^{(k)})^{-1} \mathbf{A}_k\|} = O(d_k^{-\delta_1}) = o(d_k^{-\delta_2}), \end{aligned}$$

where the first two rates are $o(d_k^{-\delta_2})$ by Assumption (R2) (or (R2P) for the rate of convergence in probability for $\mathbf{S}_{12}^{(k)}$), and the last by Assumption (M1). Hence the first r_k eigenvalues in $\mathbf{S}_{11}^{(k)}$ dominate those in $\mathbf{G} := \sum_{j=2}^3 (\mathbf{S}_{1j}^{(k)} + \mathbf{S}_{j1}^{(k)}) + \sum_{i=2}^3 \sum_{j=1}^{d_k} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top}$, and hence the r_k eigenvectors corresponding to the largest r_k eigenvalues of $\mathbf{G} + \mathbf{S}_{11}^{(k)}$ coincide with those of $\mathbf{S}_{11}^{(k)}$'s for large enough d_k , which are all necessarily in $\mathcal{N}^\perp$. This means that the $(r_k + 1)$ th largest eigenvalue of $\mathbf{G} + \mathbf{S}_{11}^{(k)}$ and beyond will have eigenvectors in $\mathcal{N}$ since $\text{rank}(\mathbf{S}_{11}^{(k)}) = r_k$. Take $\mathbf{w} \in \mathcal{N}$, then it is easy to see that

$$\mathbf{w}^\top (\mathbf{G} + \mathbf{S}_{11}^{(k)}) \mathbf{w} = \mathbf{w}^\top \left(\sum_{i=2}^3 \sum_{j=1}^{d_k} \mathbf{Q}_{ij}^{(k)} \mathbf{Q}_{ij}^{(k)\top} \right) \mathbf{w} = \mathbf{w}^\top \mathbf{R}^{(k)} \mathbf{w},$$

showing that $\lambda_j(\mathbf{G} + \mathbf{S}_{11}^{(k)}) = \lambda_j(\mathbf{R}^{(k)})$ for $j \in [d_k] \setminus [r_k]$, which is (8.30). The mode of convergence is either almost surely (with (R2) in place) or in probability (with (R2P) in place). This completes the proof of the first part of the lemma without Assumption (E3).

The proof of the lemma with Assumption (E3) is in fact completely parallel to the proof of the first part, using the corresponding results from Lemma 3 in (8.27), (8.28) and (8.29). For (8.30), the proof goes exactly the same using rates from Assumption (R2)' or (R2P)'. This completes the proof of the lemma. $\square$

Proof of Theorem 2 and 3. For $j \in [r_k]$ and $k \in [K]$, we can decompose

$$\begin{aligned}\lambda_j(\widehat{\mathbf{R}}^{(k)}) &\leq \lambda_j(\mathbf{R}^{(k)}) \left\{ 1 + \left| \frac{\lambda_j(\widehat{\mathbf{R}}^{(k)})}{\lambda_j(\mathbf{S}^{(k)})} - 1 \right| + \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| + \left| \frac{\lambda_j(\widehat{\mathbf{R}}^{(k)})}{\lambda_j(\mathbf{S}^{(k)})} - 1 \right| \cdot \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| \right\} \\ &= \lambda_j(\mathbf{R}^{(k)})(1 + O_{a.s.}(a_T(\delta_2))) \quad \text{or} \quad \lambda_j(\mathbf{R}^{(k)})(1 + O_P(a'_T(\delta_2))),\end{aligned}$$

where $a_T(\delta_2)$ or $a'_T(\delta_2)$ has the same rate as $\max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|$ (almost surely or in probability, respectively) in the first half of Lemma 4 when Assumption (E3) is not in place, or having the rate in the latter half of Lemma 4 with Assumption (E3), (R1)' and (R2)'. The last equality used $\left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| = o_{a.s.}(1)$ or $o_P(1)$ from the assumption in the theorem, the fact that $\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right| = O\left(\max_{j \in [r_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|\right)$ (almost surely or in probability) from the result of Lemma 4, and the fact that

$$\left| \frac{\lambda_j(\widehat{\mathbf{R}}^{(k)})}{\lambda_j(\mathbf{S}^{(k)})} - 1 \right| \leq \frac{\left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right|}{1 + \left(\frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\mathbf{R}^{(k)})} - 1 \right)}. \quad (8.31)$$

Also, for $j \in [d_k] \setminus [r_k]$ and $k \in [K]$,

$$\begin{aligned}\lambda_j(\widehat{\mathbf{R}}^{(k)}) &\leq \lambda_j(\mathbf{R}^{(k)}) \left(1 + \left| \frac{\lambda_j(\widehat{\mathbf{R}}^{(k)})}{\lambda_j(\mathbf{S}^{(k)})} - 1 \right| \right) + |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})| + \left| \frac{\lambda_j(\widehat{\mathbf{R}}^{(k)})}{\lambda_j(\mathbf{S}^{(k)})} - 1 \right| \cdot |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})| \\ &= \lambda_j(\mathbf{R}^{(k)}) + O_{a.s.}\left(\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}^{(k)})} - 1 \right|\right) \quad \text{or} \quad \lambda_j(\mathbf{R}^{(k)}) + O_P\left(\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}^{(k)})} - 1 \right|\right),\end{aligned}$$

where we used $\lambda_j(\mathbf{R}^{(k)}) \leq 1$ from Theorem 1, (8.31), and the fact that from Lemma 4 we see that in both scenarios with or without Assumption (E3),

$$\max_{j \in [d_k] \setminus [r_k]} |\lambda_j(\mathbf{S}^{(k)}) - \lambda_j(\mathbf{R}^{(k)})| = O\left(\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}^{(k)})} - 1 \right|\right),$$

either almost surely (with (R2) or (R2)' in place) or in probability (with (R2P) or (R2P)' in place). The rates for $\max_{j \in [d_k]} \left| \frac{\lambda_j(\mathbf{S}^{(k)})}{\lambda_j(\widehat{\mathbf{R}}^{(k)})} - 1 \right|$ from Lemma 4 then finishes the proofs for both scenarios.

With $r_k = \{j : \lambda_j(\mathbf{R}^{(k)}) > 1, j \in [d_k]\}$ from the result of Theorem 1, and the proved almost sure or in probability closeness of $\lambda_j(\widehat{\mathbf{R}}^{(k)})$ and $\lambda_j(\mathbf{R}^{(k)})$ for $j \in [d_k]$ above, we can see that then

$$\widehat{r}_k = \max\{j : \lambda_j(\widehat{\mathbf{R}}^{(k)}) > 1, j \in [d_k]\} = r_k$$

almost surely or in probability. This completes the proof Theorem 2 and 3. $\square$

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